



MATHEMATICAL MODELING ON THE DYNAMICS OF SMOKING IN
SOCIETY AMONG DIFFERENT GENDERS

BY
KADHORO BLESSED
R219098L
bkadthoro@gmail.com

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SUPERVISOR
MR M CHINYOKA

DECLARATION

I Kadhoru Blessed declare that the work submitted is my own and that appropriate credit has been given where necessary and reference has been made to the work of others

DEDICATION

In memory of those who have shaped my journey and inspired my passion for knowledge, and to my mentors, whose wisdom and insight have guided me along the way.

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ABSTRACT

This research introduces a detailed mathematical model to examine the dynamics of smoking behavior in society, with an emphasis on differences between genders. It investigates various effects of smoking, which include smoking-related diseases such as cardiovascular diseases, respiratory disorders, and cancer. Some of the smoking cessation strategies—such as nicotine replacement therapies, behavioral interventions, and public health campaigns are reviewed in this study. The findings indicate that integrated approaches, combining pharmacological support with behavioral counseling, significantly enhance cessation success rates. This study introduces a mathematical model to analyze smoking prevalence dynamics, differentiated by gender, and is structured as a system of six ordinary differential equations (ODEs). The model compartments include susceptible (S_m, S_f), infected or smoker population (I_m, I_f), and recovered or quitter population (R_m, R_f) for males and females, incorporating gender specific transmission rates (β_m, β_f), recovery rates (γ_m, γ_f), mortality rates ($\mu_m, \mu_f, \omega_m, \omega_f$), recruitment rates (σ_m, σ_f). Numerical simulations were performed using the Nonstandard Finite Difference (NSFD) scheme with step sizes $h = 1, h = 5$ and $h = 10$ where $\phi = \phi(h) + O(h^2)$. Key findings reveal that male populations dominate the epidemic due to higher transmission rates ($\beta_m = 0.05$ versus $\beta_f = 0.0005$), while female smoking prevalence declines rapidly. The basic reproduction number R_0 was derived to assess epidemic thresholds, highlighting the insignificance of female-related parameters in sustaining smoking spread. The model underscores the importance of gender-targeted interventions, with sensitivity analysis identifying male transmission (β_m) and recruitment (σ_m) rates as the most influential levers for control. Sensitivity analysis reveals the need for control measures to reduce transmission rate and recruitment rate. The suggested control measures include smoking bans, awareness campaigns, educational campaigns and cessation strategies, Youth prevention programs. Limitations include homogeneous mixing assumptions and fixed parameters, suggesting future extensions to incorporate network effects, age structure, and empirical validation. This work provides a framework for policymakers to design optimized, gender-specific anti-smoking strategies.

Contents

0.1	INTRODUCTION	8
0.2	LITERATURE REVIEW	8
0.3	MODEL FORMULATION	8
0.4	MODEL ANALYSIS	8
0.5	CONCLUSION AND RECOMMENDATIONS	8
1	Introduction	9
1.1	introduction to the chapter	9
1.2	Background of the study	9
1.3	Statement of the problem	10
1.4	Aim of the study	11
1.5	Objectives	11
1.6	Significance of the study	11
1.7	delimitations	12
1.8	Definition of terms	12
1.8.1	Mathematical model	12
1.8.2	Smoking prevalence	12
1.8.3	Ordinary Differential Equations (ODEs)	12
1.8.4	Compartmental Models	12
1.9	Summary	13
2	LITERATURE REVIEW	15
2.1	Introduction	15
2.2	Effects of smoking	15
2.2.1	Health effects	15
2.2.2	Economic effects	16
2.2.3	Social effects	17
2.3	Prevalence and patterns of smoking in Zimbabwe	17
2.4	Strategies to reduce smoking.	17
2.4.1	Smoking cessation strategies	18
2.5	Mathematical preliminaries	19
2.5.1	Ordinary differential equations	19
2.5.2	Well posedness of a solution	19
2.5.3	Existence and uniqueness of a solution	20
2.5.4	Linearization	20
2.5.5	Initial Value Problems	20
2.5.6	Stability analysis	20
2.5.7	Lyapunov function	21
2.5.8	Next Generation Matrix	22
2.5.9	Invariant Sets	23
2.6	Empirical research related to smoking	23
2.7	Summary	24

3	MODEL FORMULATION	25
3.1	Introduction	25
3.2	Model Formulation	25
3.3	Deterministic Model Formulation	25
3.4	Detailed definitions of variables	26
3.5	Biological definition of parameters	26
3.6	Model Assumptions	27
3.7	Flow diagrams to show the dynamics	28
3.8	Model equations	30
4	MODEL ANALYSIS	31
4.1	introduction	31
4.2	Invariant region	32
4.3	Existence and uniqueness of solution	32
4.3.1	Proof	32
4.3.2	Existence Theorem:	32
4.3.3	Uniqueness Theorem.	32
4.4	Positivity and Boundedness of a solution	34
4.5	Smoking Free Equilibrium point(E_0)	37
4.6	Reproductive Number R_0	37
4.7	Local Stability of Smoking Free Equilibrium	38
4.8	Global stability Analysis for E_0	39
4.9	Endemic equilibrium points	41
4.9.1	Local Stability of the Endemic Equilibrium	41
4.10	Sensitivity analysis of R_0	42
4.10.1	sensitivity indices table	44
4.11	Numerical simulations	45
5	Discussion and Conclusion	49
5.1	Key findings	49
5.2	Limitations of the study	49
5.3	Recommendations	50
5.4	Future works	51

0.1 INTRODUCTION

0.2 LITERATURE REVIEW

0.3 MODEL FORMULATION

0.4 MODEL ANALYSIS

0.5 CONCLUSION AND RECOMMENDATIONS

Chapter 1

Introduction

1.1 introduction to the chapter

This chapter will provide a comprehensive overview of the study's background, explaining its primary aims and objectives. It will thoroughly explain the major aim of the research and outline the specific objectives that guide the investigation. Furthermore, this chapter will review the statement of the problem, highlighting the significance of the study in addressing this issue. In addition to these elements, key terms that will be used in this project will be clearly defined to ensure a common understanding for readers.

1.2 Background of the study

Smoking involves inhaling and exhaling the smoke produced by burning plant leaves, with tobacco being the most commonly smoked plant. According to Sweanor and Hilton, tobacco contains nicotine, an addictive alkaloid that produces both stimulating and calming psychoactive effects [28]. Tobacco use remains an epidemic in Zimbabwe, and the government's inaction regarding this issue safeguards the tobacco industry, even as the annual death toll rises. [5]. SP Saha et al. (2007) report that the average smoking prevalence is 42% for men and 24% for women in developed countries, while in less developed countries, the figures are 48% for men and 7% for women. [28]. Smoking is addictive and can lead to numerous health issues, including lung cancer, heart disease, stroke, chronic obstructive pulmonary disease, and other respiratory conditions. Reducing smoking is crucial for managing these diseases and saving lives. Alkhudari et al. (2014), noted that tobacco use is regarded as a disease that can be transmitted through social interactions, much like the spread of infectious diseases. [1]. Smoking is a public health concern globally since its impact on public health results in many preventative deaths. According to most studies that were done, it is evident that the negative impacts of smoking can overcome its benefits. Smoking brings social, economic, physical, and environmental problems. Some of the notable economic effects of smoking include health care expenses, loss of productivity due to illness or premature death, and also tobacco smokers spending much on cigarettes, especially chain smokers. According to Atlas Zimbabwe (2019), the economic impact of smoking in Zimbabwe amounts to 17,640,177,664 Zimbabwean dollars. This figure encompasses both direct healthcare costs and indirect expenses associated with reduced productivity due to illness and early mortality. [14]. The social effects include social stigma and social isolation due to restrictions on public smoking and they can strain interpersonal relationships, especially in cases where one partner is a smoker and the other is not. Nicotine can lead to addiction which results in physical and psychological dependency on tobacco. It has been discovered that smoking kills almost half of its users if they do not quit at an early stage and also ex users can also suffer the consequences if they quit after an advanced stage. The World Health Organization states that tobacco use is the leading pre-

ventable cause of death globally, claiming over five million lives annually—surpassing the combined deaths from tuberculosis, HIV/AIDS, and malaria. Without immediate intervention, this number is projected to rise to over eight million by 2030. [21]. The implementation of the World Health Organization Framework Convention on Tobacco Control (WHO FCTC) has the potential to reverse this devastating epidemic, provided it is prioritized as a key public health issue by political leaders in countries. [9]. The WHO also emphasized the importance of developing national capabilities to implement effective and sustainable tobacco control programs, which are essential for addressing the tobacco epidemic. It is very important to study the dynamics of smoking in the society. Understanding how smoking can spread among males and females helps in drafting cessation programs knowing exactly where to start from. Mathematical models can help analyze the spread of smoking and forecast the effects of smokers on the community, aiding efforts to decrease the number of smokers. After understanding the dynamics, it is vital to introduce cessation programs and analyse their effectiveness in reducing smoking and its impacts. This paper is aimed at analysing how smoking can spread in the community. This is going to consider that the smoking behaviour of females is different from that of males hence the model is gender-based. Unlike other researchers who consider the general dynamics without considering gender differences. This paper is going to explore the impacts of educational campaigns on the dangers of smoking as a way of reducing smoking.

1.3 Statement of the problem

It is essential to incorporate both genders in the research design, implementation, analysis, and interpretation processes. Gender-based comparisons can reveal insights into smoking dynamics that may inform new treatment strategies and cessation interventions tailored to each gender. Additionally, analyzing data by gender will improve the scientific accuracy of the findings. Research indicates that men and women often exhibit distinct smoking behaviors, for instance, men may be more likely to smoke in social settings than women. By examining these patterns, we can gain insights into the underlying motivations for smoking among different genders. Moreover, the health consequences of smoking can vary between genders due to biological differences and varying health risks associated with smoking-related diseases. For example, women may experience different health outcomes from smoking compared to men, particularly in relation to reproductive health and lung disease, therefore gender models can highlight these differences and inform targeted public health interventions. Gender-specific smoking cessation programs are often more effective than the general approach since women may face unique challenges in quitting smoking, such as societal pressures and associated mental health issues. By incorporating gender differences into our mathematical modeling, we can develop specified strategies that address these specific challenges. Understanding how gender influences smoking behavior can help policymakers design more effective anti-smoking campaigns and regulations that align with the specific experiences and challenges faced by different genders. In addition, considering gender differences leads to more effective disease control strategies and improved health outcomes for diverse populations. According to McGregor et al. (2016), neglecting gender has negative implications for human health, as it not only results in missed opportunities for therapeutic advancements but also because gender can influence the frequency and severity of adverse events. [24]. Existing models focus on general dynamics without considering gender differences. These models overlook the fact that males and females behave differently and the dynamics slightly differ. This model is aimed at addressing this gap by formulating a new model which considers gender differences.

1.4 Aim of the study

The major aim of this study is to capture the dynamics of smoking behavior among different genders in society.

1.5 Objectives

The objectives of this study are;

1. To formulate a mathematical model that illustrates the dynamics of smoking behavior among different genders in society.
2. To study and analyze the dynamics represented by the model. assess the stability of the equilibrium points within the model

1.6 Significance of the study

Studying mathematical modeling on the dynamics of smoking in society among different genders is significant for several reasons.

This model can help predict trends in smoking behavior among different genders in society. Men and women often have different smoking initiation and cessation patterns influenced by social, cultural, and psychological factors. Studying these differences helps identify the specific motivations and barriers for each gender, and by understanding these dynamics, policymakers can modify interventions to specific groups more effectively. Modeling the dynamics of smoking in society can provide knowledge on the behavior of different genders in society in handling smoking, which is crucial for designing evidence-based public health policies. Such programs can address the unique challenges faced by men and women, improving their effectiveness and success rates. The health risks associated with smoking can vary by gender due to biological differences and understanding these variations is crucial for predicting health outcomes and ensuring that public health initiatives are designed to address the specific risks faced by each gender. Gender-sensitive research can guide policymakers in creating more effective anti-smoking campaigns and regulations by recognizing how gender influences smoking behavior. There is often a relationship between smoking and mental health issues, which may differ by gender. Understanding these connections can help in designing treatment approaches that address both smoking and mental health which are specific to a certain gender. The model can assist in determining the most efficient allocation of resources for educational campaigns aimed at reducing smoking. This can help maximize the impact of interventions while minimizing costs by allocating resources to the most targeted group for example if the rate of spread is greater to males than females, it is effective to target males in campaigns and other cessation interventions. By simulating the smoking dynamics in society, researchers can better understand how changes in attitudes towards smoking can influence smoking-related diseases over time. The model provides a data-driven approach to decision-making in public health. By quantifying the dynamics of smoking, policymakers can make informed choices about interventions. Modeling the dynamics of smoking allows for the evaluation of different intervention strategies. This can help identify which approaches are most effective in reducing smoking rates and related diseases

1.7 delimitations

This research will focus on the general dynamics without considering age differences in Zimbabwe. The study will focus on all age groups residing in urban and rural areas of Zimbabwe. It will specifically examine the smoking behaviors of men and women of all age groups without considering age differences. The research will employ quantitative methods and statistical analyses to gather and interpret data. Qualitative approaches, such as interviews or focus groups, will not be included in this study. The study will be conducted within the context of Zimbabwe's sociocultural and economic environment. It will focus on the impact of social attitudes towards smoking of the local Zimbabweans but will not compare smoking dynamics with other countries. The primary focus will be on developing and analysing mathematical models to describe the smoking dynamics among different genders. By defining these delimitations, the study aims to maintain a clear focus on the mathematical modeling of smoking dynamics among genders in Zimbabwe, ensuring that the findings are relevant and applicable within the specified context.

1.8 Definition of terms

In this study, some of the important terms and concepts are defined below:

1.8.1 Mathematical model

A mathematical model represents a system or process through the use of mathematical concepts and language. [6]. Mathematical modeling involves using mathematics to represent and analyze real-world problems. By translating these problems into mathematical language, one can find solutions that are then interpreted to make predictions about the real world.

1.8.2 Smoking prevalence

Refers to the proportion of a population that engages in smoking tobacco products during a specified time period. It is typically expressed as a percentage and can vary based on factors such as : demographics (Age, gender, socioeconomic status, and education), geographical location, cultural, regulatory environment and health awareness.

1.8.3 Ordinary Differential Equations (ODEs)

Ordinary Differential Equations (ODEs) are mathematical equations that involve functions of a single variable along with their derivatives. They are fundamental in modeling various phenomena in science, and engineering, and modeling the spread of infectious diseases over time. They describe how the number of infected, susceptible, and recovered individuals changes with time[16] (Wolfram Notebook).

1.8.4 Compartmental Models

Compartmental models are a very general modeling technique that is often applied to the mathematical modeling of infectious diseases [15] (Wikipedia). These models divide the population into compartments (such as susceptible, infected, and recovered) and track the flow of individuals between these compartments using differential equations.

1.9 Summary

This introductory chapter has provided an overview of the problems associated with smoking in Zimbabwe, which serves as the primary motivation for this study. Describes the research objectives, objectives and significance of this study. The chapter also highlights various effects of smoking, including health effects. The chapter also provides an insight of this research including delimitations of this study, and some terms used in this project are also defined in this chapter.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

This chapter goes into detail on the whole concept of smoking, including its social, economic, and health effects on society. Smoking-related diseases are explained in this chapter. The chapter will also review various cessation programs and the gender distribution of smoking in Zimbabwe. The chapter reviews previous studies on the dynamics of smoking. The mathematical tools that are used in the formulation and analysis of the model are also explained in this chapter.

2.2 Effects of smoking

2.2.1 Health effects

From previous research that was done, it is evident that smoking is harmful to human health. Smoking causes various respiratory diseases which result in death. Smoking is linked to various diseases, including lung cancer, heart disease, chronic obstructive pulmonary disease (COPD), and other respiratory conditions. It poses a significant public health issue, as more than 16 million Americans are currently affected by smoking-related illnesses. [7]. Smoking and passive smoking (exposure to secondhand smoke) lead to more than 480,000 deaths annually in the U.S. Smoking is said to affect almost every organ in the body and is responsible for numerous diseases.

Lung cancer

Tobacco smoke contains carbon monoxide, tar, nicotine and thousands of other chemicals, which are addictive, toxic, cancer-causing, and are harmful to the arteries which are suppliers of blood to the heart. [30]. Human bodies will repair the damage caused as a result of these chemicals, but as the damage increases, the body will be unable to manage to heal. This will result in the formation of cancer cells. Inhaling tobacco smoke can damage the alveoli, the tiny air sacs crucial for lung function, which facilitate gas exchange by transferring oxygen into the bloodstream and expelling carbon dioxide during exhalation. It is estimated that approximately 90 percent of lung cancers are linked to smoking, and tobacco use accounts for about 22 percent of cancer-related deaths. [40]. World Health Organization (WHO) stated that, lung cancer is the leading cause of cancer-related deaths and it is responsible for approximately 1.76 million fatalities globally each year.

Chronic Obstructive Pulmonary Disease (COPD)

Chronic obstructive pulmonary disease (COPD) is a progressive lung condition caused by damage to lung tissue. This damage leads to inflammation and swelling in the airways, restricting airflow in and out of the lungs. COPD is often the result of long-term exposure to smoke, fumes, dust, or chemicals, with cigarette smoke being the most common cause. COPD is most often caused by long-term exposure to smoke, fumes, dust or chemicals and the most common cause is cigarette smoke [25]. Different tobacco leaves contain varying nicotine levels, and the size of the inhaled smoke particles influences where they settle in the airways. Smoking is one of the key contributors to COPD, causing airway inflammation, structural changes, and lung damage. Findings indicate that tobacco smoke affects children's lung development if they are exposed to it, and they will be at the risk of chronic obstructive pulmonary disease (COPD) later in life.[44]. Tobacco smoking is responsible for over 70% and 30-40% of COPD cases in high-income countries and low- and middle-income countries respectively with household air pollution being another significant risk factor. [35]. Secondhand smoke exposure is a known risk factor for both the development and worsening of COPD. Studies indicate that exposure during childhood or adulthood increases the likelihood of COPD-related death as breathing in secondhand smoke can harm the respiratory system, especially in those with pre-existing conditions like asthma, infections, or allergies. Prolonged exposure may cause inflamed and narrowed airways, blockages, and damage to lung tissue. Additionally, it heightens the risk of COPD, worsens symptoms in affected individuals, and accelerates lung function decline.

Cardiovascular Diseases

Smoking is a major contributor to cardiovascular disease (CVD), accounting for one in every four CVD-related deaths. Estimates indicate that smoking increases the risk of coronary heart disease by 2 to 4 times and the risk of stroke by 2 to 4 times compared to non-smokers. [48]. It is assumed that any amount of smoke can cause cardiovascular disease. The chemicals inhaled when smoking are primary causes of damages to the heart and blood vessels which in increases the chances of developing atherosclerosis, or plaque buildup in the arteries and any smoker can be affected since any amount of smoking which include occasional smoking can cause this damage. Smoking significantly heightens the risk for certain individuals, especially women using birth control pills, those with diabetes, and others with heart disease risk factors like high blood pressure and obesity. It adversely affects the cardiovascular system by increasing triglyceride levels, lowering "good" cholesterol (HDL), and promoting blood clotting, which can block circulation. This leads to serious conditions such as atherosclerosis, coronary heart disease (CHD), heart attacks, strokes, and abdominal aortic aneurysm. Additionally, smoking raises the likelihood of developing peripheral artery disease (PAD), which increases the risk of coronary heart disease and strokes. [10]

2.2.2 Economic effects

The health effects of smoking are associated with economic problems which include high medical costs in trying to cure smoking-related diseases. Economic issues include the high medical costs associated with treating smoking-related diseases. In the United States, the costs of heart disease, stroke healthcare expenses and lost of productivity were estimated at \$351 billion in 2003.[28]. According to the Tobacco Atlas, the cost of smoking in Zimbabwe in 2019 including direct costs related to healthcare expenses and indirect costs related to lost productivity caused by illness and premature death was 17640177664 Zimbabwean dollars. [14].

2.2.3 Social effects

When smoking becomes addictive, it will strain interpersonal relationships between smokers and nonsmokers. Passive smoking is another social effect that may ruin interpersonal relationships at the family level. Smoking may also lead to social stigma and isolation of smokers due to public smoking restrictions.

2.3 Prevalence and patterns of smoking in Zimbabwe

According to the Tobacco Atlas, tobacco smoking is a pandemic in Zimbabwe that needs to be addressed as the death toll due to tobacco-related diseases rises. According to the World Health Organization, underage smoking is increasing in Zimbabwe. While the latest statistics are not surprising for the region, many in Zimbabwe worry that tobacco use may be a cover for more serious underlying problems. [23]. In Zimbabwe, 8372 people die each year due to tobacco smoking. There is a slight downward trend in current smoking prevalence in the general population in Zimbabwe.

Smoking statistics in Zimbabwe according to the Global State of Tobacco Harm Reduction [31]

TABLE 1

description	number of people	reference
Number of smokers	757.1 thousand	[39]
Adult tobacco smoking prevalence (Females)	0.7%	[43]
Adult tobacco smoking prevalence (males)	18%	[43]
Number of daily smokers	1.1 million	[45]
Adult daily tobacco smoking prevalence	11.8%	[45]
Adult daily tobacco smoking prevalence (females)	1.2%	[45]
Adult daily tobacco smoking prevalence (Males)	22.9%	[45]
Annual death	8.7 thousand	[11]
Annual death (Females)	2 thousand	[11]
Annual death (Males)	6.7 thousand	[11]
Annual % of all deaths due to tobacco smoking	4.51%	[11]
Annual % of all deaths due to tobacco smoking (female)	2.28%	[11]
Annual % of all deaths due to tobacco smoking (male)	6.42%	[11]

2.4 Strategies to reduce smoking.

Educational campaigns and awareness are one of the key strategies for reducing smoking. Some people end up smoking because they are not aware of the consequences of smoking hence educating them can help. In addition, some people want to quit but they don't know how, therefore educating them on several quitting strategies is of greater importance. Sadeghi et al. state that anti-smoking campaigns are vital components of comprehensive tobacco control programs. These campaigns aim to educate the public about the dangers of smoking, alter attitudes and beliefs towards tobacco use, enhance intentions and attempts to quit, and encourage cessation [27]. Educational campaigns are very crucial in reducing new infections and in encouraging quitting to those who are smoking. There

are cessation strategies that can be done when the smoker decides to quit. Hersi et al., 2019 suggest that quitting smoking is not easy since nicotine is addictive therefore there is need for these strategies. These include approved pharmacotherapies where Nicotine replacement therapy (NRT) and varenicline is used and behavioural therapies [8]. The WHO Report on the Global Tobacco Epidemic 2021 indicates that countries are still adopting effective policies to reduce tobacco use, despite the challenges posed by the global pandemic. [45].

2.4.1 Smoking cessation strategies

As highlighted by Hersi et al., 2019, it is difficult for individuals to quit smoking since nicotine is addictive therefore the following strategies may be recommended :

Varenicline

Varenicline (known as Champix or Chantix) is commonly prescribed by doctors to assist individuals in quitting smoking and it releases dopamine, a brain chemical associated with pleasure. While nicotine from smoking also elevates dopamine levels, it does so more intensely but for a brief period, resulting in cravings for cigarettes. Varenicline helps by blocking nicotine receptors in the brain, which lessens cravings and withdrawal symptoms which are usually experienced by people attempting to quit smoking. To help prevent mood declines, doctors often recommend or prescribe nicotine patches or antidepressants alongside varenicline. Current guidelines from the American Thoracic Society advocate using this medication for at least 12 weeks as a first-line approach for smoking cessation, favoring it over starting with nicotine patches, e-cigarettes, or other treatments [32], however, a doctor may suggest combining it with nicotine patches.

Nicotine replacement therapy

Nicotine replacement therapy (NRT) helps quitters to handle withdrawal symptoms when quitting smoking by delivering a controlled dose of nicotine without the harmful chemicals found in tobacco smoke. It is available in several forms, such as gum, patches, sprays, inhalers, and lozenges. Patches, which can be purchased in pharmacies, release nicotine gradually through the skin, reducing symptoms like cravings, nausea, and mood swings. As individuals transition to lower-dose patches, the nicotine dosage decreases over time, helping to eliminate cravings. Nicotine patches can be used differently with some individuals use them continuously for 24-hour coverage, while others prefer to take them off at night. Usually, doctors suggest the most suitable method. In contrast, inhalers, gum, lozenges, and sprays offer quick relief for severe cravings, though their effects are short-lived. Nicotine replacement therapies (NRTs) are vital for helping individuals quit smoking. Options include the nicotine patch, which releases nicotine over 16 to 24 hours; nicotine gum, providing quick relief; nicotine lozenges, which dissolve gradually; nicotine inhalers, simulating the act of smoking; and nicotine nasal spray, offering rapid absorption. These therapies cater to various preferences, aiding in nicotine addiction management.[12]

Bupropion

Bupropion (known as Zyban) is an antidepressant that can aid individuals in quitting smoking. Similar to varenicline, it helps counteract the dopamine deficiency associated with nicotine withdrawal. This medication may also reduce irritability and concentration challenges that people often face when they stop smoking.

Cognitive behavioral therapy (CBT)

Cognitive behavioral therapy (CBT) is a counseling approach designed to help people change unhelpful habits. In 2016, researchers launched a clinical trial to explore how cognitive behavioral interventions could benefit adults over 18 who smoke at least eight cigarettes a day and wish to quit. They found that a specific treatment called cognitive behavioral smoking cessation treatment with behavioral activation (SCBSCT-BA) not only improved the chances of quitting but also reduced the likelihood of relapse and helped manage mood swings related to the quitting process. Additionally, a 2008 study involving 304 adult smokers who participated in 20 weeks of CBT focused on strategies to prevent smoking. The results indicated that this approach could support long-term abstinence from smoking. The effectiveness of CBT over 20 weeks indicates that focusing on developing cognitive and behavioral strategies for sustaining non-smoking during a prolonged treatment period may assist smokers in maintaining abstinence in the long run. Currently, the minimum length of therapy remains unclear [40].

2.5 Mathematical preliminaries

2.5.1 Ordinary differential equations

Differential equations are essential mathematical instruments used for modeling across numerous disciplines, such as science, engineering, economics, mathematics, physics, aeronautics, astronomy, dynamics, biology, chemistry, medicine, environmental sciences, social sciences, banking, and more. An ordinary differential equation (ODE) is a specific kind of differential equation that includes only one independent variable, with all derivatives pertaining to that variable. Time (t) is the most common variable used and other variables can also be applied in certain cases. Examples include:

1.

$$y^n = F(x, y, y', y'', y''', \dots, y^{n-1})$$

2.

$$\frac{dy}{dt} = 6t - 3y$$

A solution of an Ode

Solutions to ordinary differential equations are functions that satisfy the equation and fulfill any specified initial conditions, holding true for all values of the independent variable within a defined domain and they can be obtained either analytically or numerically, depending on the equation's complexity and the techniques available.

2.5.2 Well posedness of a solution

An Initial value problem (IVP) is considered well-posed if it satisfies three properties:

1. existence of at least one solution,
2. uniqueness of solutions, and
3. stability of solutions with respect to initial/boundary conditions.

2.5.3 Existence and uniqueness of a solution

Existence Theorem

Theorem 1

a. If f is continuous on an open rectangle

$$R : a < x < b, c < y < d$$

that contains

$$(x_0, y_0)$$

then the initial value problem

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

has at least one solution on some open subinterval of (a, b) that contains x_0

b. If both f and f_y are continuous on R then Equation has a unique solution on some open subinterval of (a, b) that contains x_0

2.5.4 Linearization

Linearization refers to the process of transforming a nonlinear equation into a linear form. By representing the equation as a straight line, it becomes easier to analyze, solve, and draw inferences from the relationship between variables. This technique is widely employed in mathematical modeling to approximate complex phenomena and make them more manageable for mathematical treatment.

The linearization of the autonomous system $y' = f(y)$ at the critical point y_c is the linear differential system for the unknown function given by $\zeta' = f'(y_c)\zeta$

2.5.5 Initial Value Problems

An initial value problem (IVP) is a differential equation problem where the solution satisfy a given initial condition. A typical first-order IVP is of the form:

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

where: $\frac{dy}{dx}$ is the derivative of y with respect to x ,

$f(x, y)$ is a given function and

$y(x_0) = y_0$ is the initial condition, specifying the value of y at x_0

2.5.6 Stability analysis

In mathematics, stability theory focuses on the stability of solutions of differential equations and of trajectories of dynamical systems under small perturbations of initial conditions.

An equilibrium solution f_e to an autonomous system of first order ordinary differential equations is

called:

a. stable if for every $\epsilon > 0$, there exists a $\delta > 0$ such that any solution $f(t)$ having initial conditions within distance δ i.e. $\|f(t_0) - f_e\| < \delta$ of the equilibrium remains within distance ϵ i.e. $\|f(t) - f_e\| < \epsilon$ for all $t \geq t_0$

b. asymptotically stable if it is stable and, in addition, there exists $\delta_0 > 0$ such that whenever $\|f(t_0) - f_e\| < \delta_0$ then $f(t) \rightarrow f_e$ as $t \rightarrow \infty$

2.5.7 Lyapunov function

In the study of ordinary differential equations (ODEs), Lyapunov functions are scalar functions that can be utilized to demonstrate the stability of an equilibrium point and they play a crucial role in the stability theory of dynamical systems and control theory. A Lyapunov function for an autonomous dynamical system

$$\begin{cases} g : \mathbb{R}^n \rightarrow \mathbb{R}^n \\ \dot{y} = g(y) \end{cases}$$

with an equilibrium point at $y = 0$ is a scalar function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ that is continuous, has continuous first derivatives, is strictly positive for $y \neq 0$, and for which the time derivative $\dot{V} = \nabla V \cdot g$ is non positive (these conditions are required on some region containing the origin). The (stronger) condition that $-\nabla V \cdot g$ is strictly positive for $y \neq 0$ is sometimes stated as $-\nabla V \cdot g$ is locally positive definite, or $\nabla V \cdot g$ is locally negative definite

Basic Lyapunov theorems for autonomous systems

Let $x^* = 0$ be an equilibrium point of the autonomous system

$$\dot{x} = f(x).$$

and use the notation $\dot{V}(x)$ to denote the time derivative of the Lyapunov-candidate-function

$$V : (\dot{V}(x) = \frac{d}{dt}V(x(t)) = \frac{\partial V}{\partial x} \cdot \frac{dx}{dt} = \nabla V \cdot \dot{x} = \nabla V \cdot f(x).$$

1. Locally asymptotically stable equilibrium

If the equilibrium point is isolated, the Lyapunov-candidate-function V is locally positive definite, and the time derivative of the Lyapunov-candidate-function is locally negative definite:

$$\dot{V}(x) < 0 \quad \forall x \in \mathcal{B}(0) \setminus \{0\},$$

for some neighborhood $\mathcal{B}(0)$ of origin, then the equilibrium is proven to be locally asymptotically stable.

Stable equilibrium

If V is a Lyapunov function, then the equilibrium is Lyapunov stable. The converse is also true, and was proved by José Luis Massera.

Globally asymptotically stable equilibrium

If the Lyapunov-candidate-function V is globally positive definite, radially unbounded, the equilibrium isolated and the time derivative of the Lyapunov-candidate-function is globally negative definite:

$$\dot{V}(x) < 0 \quad \forall x \in \mathbb{R}^n \setminus \{0\},$$

then the equilibrium is proven to be globally asymptotically stable.

The Lyapunov-candidate function $V(x)$ is radially unbounded if

$$\|x\| \rightarrow \infty \Rightarrow V(x) \rightarrow \infty.$$

(This is also referred to as norm-coercivity.)

Example

Consider the following differential equation on

$$\mathbb{R}$$

: $\dot{x} = -x$. Considering that x^2 is always positive around the origin it is a natural candidate to be a Lyapunov function to help us study x . So let $V(x) = x^2$ on $\mathbb{R}$. Then,

$$\dot{V}(x) = V'(x)\dot{x} = 2x \cdot (-x) = -2x^2 < 0.$$

This demonstrates that the differential equation, x , is asymptotically stable around the origin and one can also establish that the equilibrium is globally asymptotically stable.

2.5.8 Next Generation Matrix

The Next Generation Matrix (NGM) method is a mathematical approach used in epidemiology to analyze the spread of infectious diseases. It helps determine key epidemic characteristics, such as the basic reproduction number (R_0), which indicates the average number of secondary infections produced by a single infected individual in a fully susceptible population.

Constructing the Next Generation Matrix

The Next Generation Matrix is derived from a system of ordinary differential equations (ODEs) describing the disease dynamics. It follows these steps:

1. Define the Disease Model

Consider a system of equations that describes the dynamics of infected compartments (e.g., exposed, infected, recovered). A general system can be written as:

$$\frac{dx}{dt} = F(X) - V(X)$$

where: $F(X)$ represents the rate of appearance of new infections in each infected compartment. $V(X)$ consists of transition terms (recoveries, deaths, etc.).

2. Compute the Jacobian Matrices F and V

where,

Matrix F represents the new infection terms and

Matrix V represents the transition terms (movement between compartments, recovery, etc.) These matrices are evaluated at the disease-free equilibrium (DFE) where no infections exist.

3. Define the Next Generation Matrix

The Next Generation Matrix is given by:

$$K = FV^{-1}$$

where: F is the Jacobian matrix of $F(X)$ at the Disease free equilibrium and,

V is the Jacobian matrix of $V(X)$ at the Disease free equilibrium

2.5.9 Invariant Sets

An invariant set is a subset of a dynamical system's state space such that if the system starts in this set, it remains in this set for all future times. Mathematically, if S is an invariant set and

$$\phi(t, x)$$

is the flow of the dynamical system, then: $x(0) \in S \Rightarrow \phi(t, x(0)) \in S \forall t \geq 0$

2.6 Empirical research related to smoking

There are many existing models that were focusing on the mathematical modeling on the dynamics of smoking using non-linear ordinary differential equations. Mathematical modeling of smoking dynamics can be effectively approached using non-linear ordinary differential equations. Castillo-Garsow et al [4] projected a straightforward mathematical model based on the PSQ framework. In their model, the total population was categorized into three groups: potential smokers, current smokers, and individuals who have permanently quit smoking.. In their study, Carlos Castillo-Garsow et al provides a general epidemiological model to capture the dynamics of drug use among adolescents, focusing on tobacco consumption. The model further refined by incorporating additional factors known to influence the increasing trend of tobacco use including peer pressure, relapse, counseling, and treatment. A notable finding from this study was that in the model where recovery efforts were influenced by peer cultures, the basic reproductive number remained constant and was solely determined by the intrinsic effectiveness of the recovery efforts. O. Sharomi and A.B. Gumel [26] presents a thorough mathematical analysis of smoking dynamics and its public health implications within a community. Their study provides a comprehensive mathematical analysis of smoking dynamics and its public health implications within a community. Building on the work of F. Brauer and C. Castillo-Chavez, their foundational model categorizes the population into non-smokers, smokers, and individuals who have quit smoking, either temporarily or permanently. A key public health finding is that the number of smokers can be effectively managed or eliminated at steady-state if the threshold remains below

one. Additionally, Erturk et al. (2012) further examined this model numerically by incorporating fractional derivatives.[36] Research by Verma and Bhadauria, (2019) titled Global dynamics of a mathematical model on smoking: impact of anti-smoking campaign which was done in India have shown that smokers population decreases as educational campaigns on potential smokers increases [34] . They also found that when individuals are educated about the fatal consequences of smoking-related diseases, they are more likely to refrain from smoking in the future, leading to an increase in the number of individuals moving to the quitter class. In a similar research, Anupama Sharma, A.K. Misra investigated smoking cessation model with media campaigns and they managed to show that when media campaigns focusing on cessation is enhanced the reproduction number is suppressed . Another research was done by Gul Zaman in 2009 aimed at analysing the asymptotic behavior of a mathematical model designed for quitting smoking. They focused on developing and examining the model while considering a compartment for occasional smokers in the quitting process [47].

Moreover, another research was done by Nur Ilmayasintal, Elly Anjarsari and Moh W. Ahdi [17] titled optimal control for smoking epidemic model in 2021. They used a Mathematical model of smoking epidemic by Gul Zaman et al and they provide four optimal control measures namely education campaign about cigarettes, vaccines, treatment and rehabilitation. From the numerical simulations carried out they showed that the number of potential smokers decreases and the number of ex-smokers increase optimally. This was sufficient to show that the system gets the optimal solution according to the values of control measures. In 2021 Omar Khyar, Jaouad Danane, and Karam Allali [20] on Mathematical Analysis and Optimal Control of Giving up the Smoking Model where they classified the population into potential smokers, occasional smokers, temporary quitters, and permanent quitters .They introduced to control measures that reduce the contact rate between smokers and nonsmokers and prevent occasional smokers from becoming chain smokers, Numerical simulation results confirm the importance of the control measures in reducing both occasional smokers and chain smokers. From the previous researches that were done, most of the researchers overlooked gender differences. They didn't take into consideration that smoking spread differently among males and females. From the data in Table 1 above it is noted that the smoking dynamics are based on gender and males and females behave differently towards smoking. Therefore, the purpose of this study is to model the dynamics of smoking in different genders.

2.7 Summary

This chapter conducts an extensive review of the literature concerning the modeling of the dynamics of smoking. It explores the impacts of smoking to individuals and investigate various strategies and measures to reduce smoking and explains various strategies implemented for individuals to quit smoking. This chapter also provides some statistics of smoking prevalences and deaths in Zimbabwe. Additionally, the chapter discusses the strengths and limitations of existing models, identifies gaps in knowledge. The chapter also explains various mathematical tools used in formulation and analysis of the model.

Chapter 3

MODEL FORMULATION

3.1 Introduction

In this section, we introduce a comprehensive mathematical model designed to study the dynamics of smoking among different genders. Intervention strategies such as smoking cessation and awareness are introduced to design a new model which consider the intervention strategies to reduce smoking. The model structure is explained, including its assumptions, compartments, and governing equations. Following this, a series of analyses to understand the model's behavior is conducted in this chapter. Stability analysis will help determine the conditions for stable smoking-free and endemic states. Sensitivity analysis will identify which parameters most influence the model's outcomes, highlighting critical factors for smoking control.

3.2 Model Formulation

A compartmental model of smoking dynamics is introduced for a population, with the population size at a given time t denoted by $N(t)$. This model divides the entire population into six groups or classes based on their epidemiological status and gender, as detailed in Table 2. The dynamics of this population can be described by the following equation, where $N(t)$ represents the total population size at time t :

$$N(t) = S_m(t) + s_f(t) + I_m(t) + I_f(t) + R_m(t) + R_f(t).$$

This equation represents the total number of individuals across all six groups at any given time t . As individuals transition between these compartments due to factors such as smoking initiation, cessation, or death, the total population size $N(t)$ changes dynamically over time.

3.3 Deterministic Model Formulation

We develop a deterministic compartmental model to analyze the dynamics of smoking among various genders within a population. This model categorizes the population into specific compartments according to their epidemiological status: Susceptible males (S_m), Susceptible females (S_f) susceptible females, (I_f) female smokers, (I_m) male smokers, male quitters (R_m), and female quitters (R_f). The population dynamics can be represented by the following equation, where $N(t)$ denotes the total population size at time t :

$$N(t) = S_m(t) + S_f(t) + I_m(t) + I_f(t) + R_m(t) + R_f(t).$$

3.4 Detailed definitions of variables

The total population at time t is divided into six compartments, each representing different groups based on their smoking status and gender which are susceptible males (non-smokers) S_m , Susceptible females (non-smokers) S_f , male smokers I_m , female smokers I_f , male quitters R_m and female quitters R_f . Susceptible males (S_m) represent males who have not started smoking and are susceptible to smoking. They remain at risk of smoking if exposed to smoking individuals (both males and females), leading to potential transmission. Susceptible females (S_f) represent females who have not started smoking and are susceptible to smoking. They remain at risk of smoking if exposed to smoking individuals (both males and females), leading to potential transmission. Male smokers I_m represent males who move from susceptible males and start smoking. They are smokers and if they interact with the susceptible individuals, they can cause them to smoke. Female smokers I_f represent females who move from susceptible females and they start smoking. Like Infected males, they are smokers and if they interact with the susceptible individuals, they can cause them to smoke. Male quitters (R_m) represent males who quit smoking while R_f (female quitters) represent females who quit smoking. The variables used in this model formulation are summarised in Table 3.1:

Table 3.1

Variable	description
S_m	Susceptible males (non-smokers)
S_f	susceptible females (non-smokers)
I_m	male smokers(active smokers)
I_f	female smokers (active smokers)
R_m	male quitters
R_f	female quitters

3.5 Biological definition of parameters

The parameters are described in Table 3

Table 3.2

parameter	description	value[ref]
β_m	rate at which individuals become smokers due to interaction with male smokers	0.05 [18]
β_f	rate at which individuals become smokers due to interaction with female smokers	0.0005[estimate]
γ_m	rate at which males quit smoking	0.0031 [47]
γ_f	rate at which females quit smoking	0.0031[estimate]
μ_m	natural death rate for males	0,04288 [19]
μ_f	natural death rate for females	0,0415744 [19]
ω_f	female death rate due to smoking-related diseases	0.0228 [11]
ω_m	male death rate due to smoking-related diseases	0.0642 [11]
σ_m	recruitment rate of males	0.02[estimate]
σ_f	recruitment rate of females	0.002[estimate]
ϵ_m	rate at which males relapse	0.06[18]
ϵ_f	rate at which females relapse	0,0006[estimate]

3.6 Model Assumptions

Several assumptions are made to simplify the modeling process and provide a basis for analysis:

1. The total population at time t is divided into six compartments which are susceptible males (non-smokers) S_m , Susceptible females (non-smokers) S_f , male smokers I_m , female smokers I_f , male quitters R_m and female quitters R_f . There is no movement of individuals into or out of the population during the simulation period and the population will remain constant at $N(t)$ which is given by

$$N(t) = S_m(t) + S_f(t) + I_m(t) + I_f(t) + R_m(t) + R_f(t).$$

2. Smoking transmission occurs after interaction with infected individuals and smoking is transmitted at the rates β_m and β_f where β_m is rate at which individuals become smokers due to interaction with male smokers and β_f is rate at which individuals become smokers by interacting with female smokers.
3. The rate of change of potential smokers at time t is given by $\frac{dS_m}{dt} + \frac{dS_f}{dt}$ where

$$\frac{dS_m}{dt} = \delta_m - \beta_m S_m I_m - \beta_f S_m I_f - \mu_m S_m$$

and

$$\frac{dS_f}{dt} = \delta_f - \beta_m S_f I_m - \beta_f S_f I_f - \mu_f S_f$$

while δ_m is recruitment rate of males, β_m is rate at which individuals become smokers due to interaction with male smokers, β_f is rate at which individuals become smokers due to interaction with female smokers, δ_f is the recruitment rate for females and μ_m and μ_f are the natural death rates for males and females respectively.

4. The population is assumed to mix uniformly between both males and females, meaning that every individual has an equal chance of encountering and interacting with every other individual in the population. Interaction between potential smokers (S_m and S_f) and smokers (I_m and I_f) may result in an increase in number of smokers. Potential smokers are recruited to the smoking population at the rate $\beta_m + \beta_f$ and an increase in number of active smokers is given by $\beta_m S_m + \beta_f S_f$. The rate of change of active smokers at time t is given by $\frac{dI_m}{dt} + \frac{dI_f}{dt}$ where

$$\frac{dI_m}{dt} = \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m$$

and

$$\frac{dI_f}{dt} = \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f$$

5. There is natural death within the population in all compartments and the natural death rate is given by μ_m for males and μ_f for females and their total number is given by

$$\mu_m(S_m + I_m + R_m) + \mu_f(S_f + I_f + R_f)$$

6. Individual smokers and quitters who have been affected may experience smoking-related disease mortality and are dying at the rate ω_m for males and ω_f for females. Their total number is given by $(\omega_m I_m + \omega_f I_f + \omega_m R_m + \omega_f R_f)$.
7. Individuals who recover from smoking may relapse to smoking again. These males and females are relapsing at the rate ϵ_m and ϵ_f respectively and their total number is given by $\epsilon_m R_m + \epsilon_f R_f$.

8. All parameters used in this model, including transmission rates, recovery rates, and death rates, are assumed to be positive values.
9. Active male and female smokers can quit smoking at the rate γ_m and γ_f respectively. The total number of people who quit smoking is given by $\gamma_m I_m + \gamma_f I_f$. The rate of change of quitters at time t is given by $\frac{dR_m}{dt} + \frac{dR_f}{dt}$ where

$$\frac{dR_m}{dt} = \gamma_m I_m - (\mu_m + \omega_m + \epsilon_m) R_m$$

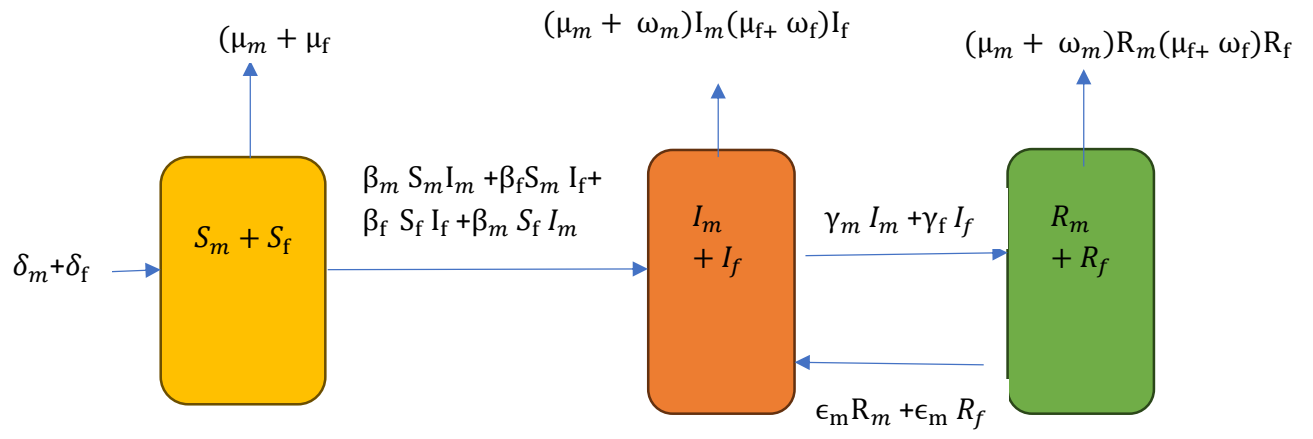
and

$$\frac{dR_f}{dt} = \gamma_f I_f - (\mu_f + \omega_f + \epsilon_f) R_f$$

These assumptions provide a framework for the deterministic model of respiratory diseases, allowing for the analysis of disease transmission dynamics and the evaluation of intervention strategies.

3.7 Flow diagrams to show the dynamics

The flow diagram below illustrates the key assumptions underlying our mathematical modeling of the dynamics of smoking behavior in society. This visual representation provides a clear overview of the relationships between various factors. By mapping out these assumptions, we can better understand the complex interactions and establish a framework for analyzing smoking patterns among different genders in Zimbabwe.



3.8 Model equations

Combining all the above assumptions leads to the following system of ordinary differential equations. These equations are derived from the key assumptions outlined in the flow diagram and are designed to capture the relationships between various factors influencing smoking behavior among different genders. These equations will serve as the basis for our analysis and simulation of smoking trends in Zimbabwe.

$$\frac{dS_m}{dt} = \delta_m - \beta_m S_m I_m - \beta_f S_m I_f - \mu_m S_m \quad (1)$$

$$\frac{dS_f}{dt} = \delta_f - \beta_m S_f I_m - \beta_f S_f I_f - \mu_f S_f \quad (2)$$

$$\frac{dI_m}{dt} = \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m \quad (3)$$

$$\frac{dI_f}{dt} = \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f \quad (4)$$

$$\frac{dR_m}{dt} = \gamma_m I_m - (\mu_m + \omega_m + \epsilon_m) R_m \quad (5)$$

$$\frac{dR_f}{dt} = \gamma_f I_f - (\mu_f + \omega_f + \epsilon_f) R_f \quad (6)$$

The initial conditions for the differential equations

$$S_m(0) = S_{m0} \geq 0$$

$$S_f(0) = S_{f0} \geq 0$$

$$I_m(0) = S_{m0} \geq 0$$

$$I_f(0) = S_{f0} \geq 0$$

$$R_m(0) = R_{m0} \geq 0$$

$$R_f(0) = R_{f0} \geq 0$$

Chapter 4

MODEL ANALYSIS

4.1 introduction

In this chapter, we will thoroughly investigate several critical aspects of our mathematical model, focusing on the existence and positivity of solutions. We will analyse equilibrium points to understand the conditions under which the system stabilises. Additionally, we will calculate the reproductive number, a vital parameter for assessing the potential spread of the smoking. We will further explore both local and global stability analyses to determine the stability of the equilibrium points under various conditions. Special attention will be given to endemic equilibrium points and their local stability, which play a crucial role in understanding long-term of smoking dynamics. We will also study sensitivity of the parameters used in this model. Finally, we will present numerical simulations to illustrate our findings and validate the theoretical results, providing a comprehensive overview of the model's behavior and implications.

$$\frac{dS_m}{dt} = \delta_m - \beta_m S_m I_m - \beta_f S_m I_f - \mu_m S_m \quad (1)$$

$$\frac{dS_f}{dt} = \delta_f - \beta_m S_f I_m - \beta_f S_f I_f - \mu_f S_f \quad (2)$$

$$\frac{dI_m}{dt} = \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m \quad (3)$$

$$\frac{dI_f}{dt} = \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f \quad (4)$$

$$\frac{dR_m}{dt} = \gamma_m I_m - (\mu_m + \omega_m + \epsilon_m) R_m \quad (5)$$

$$\frac{dR_f}{dt} = \gamma_f I_f - (\mu_f + \omega_f + \epsilon_f) R_f \quad (6)$$

The initial conditions for the differential equations

$$S_m(0) = S_{m0} \geq 0$$

$$S_f(0) = S_{f0} \geq 0$$

$$I_m(0) = S_{m0} \geq 0$$

$$I_f(0) = S_{f0} \geq 0$$

$$R_m(0) = R_{m0} \geq 0$$

$$R_f(0) = R_{f0} \geq 0$$

4.2 Invariant region

It can be observed that all the rates in the model are non-negative on the bounding plane R_6^+ . Consequently, it is straightforward to demonstrate that this region is both positively invariant and attracting. The region for the system described by the above equations can be represented as $\Pi = (S_m, S_f, I_m, I_f, R_m, R_f) \subset R_6^+ : N \leq \frac{\sigma_m}{\mu_m} + \frac{\sigma_f}{\mu_f}$. Therefore, this closed set constitutes a positively invariant and attracting region concerning the mentioned system.

4.3 Existence and uniqueness of solution

The equation represents an autonomous system with f defined as shown in the equation below. Therefore, we have

$$\begin{aligned} f_1(S_m, S_f, I_m, I_f, R_m, R_f) &= \delta_m - \beta_m S_m I_m - \beta_f S_m I_f - \mu_m S_m \\ f_2(S_m, S_f, I_m, I_f, R_m, R_f) &= \delta_f - \beta_m S_f I_m - \beta_f S_f I_f - \mu_f S_f \\ f_3(S_m, S_f, I_m, I_f, R_m, R_f) &= \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m \\ f_4(S_m, S_f, I_m, I_f, R_m, R_f) &= \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f \\ f_5(S_m, S_f, I_m, I_f, R_m, R_f) &= \gamma_m I_m - (\mu_m + \omega_m + \epsilon_m) R_m \\ f_6(S_m, S_f, I_m, I_f, R_m, R_f) &= \gamma_f I_f - (\mu_f + \omega_f + \epsilon_f) R_f \end{aligned}$$

We apply the following theorem to demonstrate the existence and uniqueness of a solution for the model. To support this proof, we require the following standard results.

Theorem

Existence and Uniqueness Theorem Define the domain D such that $D = \{t | t_0 \leq t \leq a, \|X - X_0\| \leq b\}$ Where $X = (S_m, S_f, I_m, I_f, R_m, R_f)^T$ and $X_0 = (S_{m0}, S_{f0}, I_{m0}, I_{f0}, R_{f0}, R_{m0})$ Then, the problem written in matrix form $\dot{X} = f(X)$ has a unique solution in D if f is continuous in D and there exist constants $K_1, K_2, K_3, K_4, K_5, K_6$ such that

$$|\frac{dS_m}{dt}| < K_1, |\frac{dS_f}{dt}| < K_2, |\frac{dI_m}{dt}| < K_3, |\frac{dI_f}{dt}| < K_4, |\frac{dR_m}{dt}| < K_5, |\frac{dR_f}{dt}| < K_6,$$

4.3.1 Proof

Our goal is to establish the existence and uniqueness of the solution to the system.

4.3.2 Existence Theorem:

It is clear that the functions $f_1, \dots, f_6$ as defined in are continuous and bounded in D . Thus, a solution exists in D

4.3.3 Uniqueness Theorem.

To demonstrate the uniqueness of the solution and establish that there is a unique solution to the system of equations, we will show that: $\frac{\partial f_i}{\partial x_j}, i, j = 1, 2, \dots, 6$ are continuous and bounded. We examine the following partial derivatives for the system of equations in the model. Starting with the first equation, we have:

$$\begin{aligned}\frac{\partial f_1}{\partial S_m} &= -\beta_m I_m \beta_f I_f - \mu_m, \left| \frac{\partial f_1}{\partial S_m} \right| \leq K < \infty, & \frac{\partial f_1}{\partial S_f} &= 0, \left| \frac{\partial f_1}{\partial S_f} \right| \leq K < \infty, \\ \frac{\partial f_1}{\partial I_m} &= \beta_m S_m, \left| \frac{\partial f_1}{\partial I_m} \right| \leq K < \infty, & \frac{\partial f_1}{\partial I_f} &= \beta_f S_m, \left| \frac{\partial f_1}{\partial I_f} \right| \leq K < \infty, \\ \frac{\partial f_1}{\partial R_m} &= 0, \left| \frac{\partial f_1}{\partial R_m} \right| \leq K < \infty, & \frac{\partial f_1}{\partial R_f} &= 0, \left| \frac{\partial f_1}{\partial R_f} \right| \leq K < \infty,\end{aligned}$$

Similarly, by differentiating the function f_2 in ,we have:

$$\begin{aligned}\frac{\partial f_2}{\partial S_m} &= 0, \left| \frac{\partial f_2}{\partial S_m} \right| \leq K < \infty, & \frac{\partial f_2}{\partial S_f} &= -\beta_m I_m \beta_f I_f - \mu_f, \left| \frac{\partial f_2}{\partial S_f} \right| \leq K < \infty, \\ \frac{\partial f_2}{\partial I_m} &= \beta_m S_f, \left| \frac{\partial f_2}{\partial I_m} \right| \leq K < \infty, & \frac{\partial f_2}{\partial I_f} &= \beta_f S_f, \left| \frac{\partial f_2}{\partial I_f} \right| \leq K < \infty, \\ \frac{\partial f_2}{\partial R_m} &= 0, \left| \frac{\partial f_2}{\partial R_m} \right| \leq K < \infty, & \frac{\partial f_2}{\partial R_f} &= 0, \left| \frac{\partial f_2}{\partial R_f} \right| \leq K < \infty,\end{aligned}$$

If we differentiate the third equation ,we obtain:

$$\begin{aligned}\frac{\partial f_3}{\partial S_m} &= \beta_m I_m + \beta_f I_f, \left| \frac{\partial f_3}{\partial S_m} \right| \leq K < \infty, & \frac{\partial f_3}{\partial S_f} &= 0, \left| \frac{\partial f_3}{\partial S_f} \right| \leq K < \infty, \\ \frac{\partial f_3}{\partial I_m} &= \beta_m S_m - m_1, \left| \frac{\partial f_3}{\partial I_m} \right| \leq K < \infty, & \frac{\partial f_3}{\partial I_f} &= \beta_f S_m, \left| \frac{\partial f_3}{\partial I_f} \right| \leq K < \infty, \\ \frac{\partial f_3}{\partial R_m} &= \epsilon_m, \left| \frac{\partial f_3}{\partial R_m} \right| \leq K < \infty, & \frac{\partial f_3}{\partial R_f} &= 0, \left| \frac{\partial f_3}{\partial R_f} \right| \leq K < \infty,\end{aligned}$$

Differentiating f_4 we obtain:

$$\begin{aligned}\frac{\partial f_4}{\partial S_m} &= 0, \left| \frac{\partial f_4}{\partial S_m} \right| \leq K < \infty, & \frac{\partial f_4}{\partial S_f} &= \beta_f I_f + \beta_m I_m, \left| \frac{\partial f_4}{\partial S_f} \right| \leq K < \infty, \\ \frac{\partial f_4}{\partial I_m} &= \beta_m S_f, \left| \frac{\partial f_4}{\partial I_m} \right| \leq K < \infty, & \frac{\partial f_4}{\partial I_f} &= \beta_f S_f - m_2, \left| \frac{\partial f_4}{\partial I_f} \right| \leq K < \infty, \\ \frac{\partial f_4}{\partial R_m} &= 0, \left| \frac{\partial f_4}{\partial R_m} \right| \leq K < \infty, & \frac{\partial f_4}{\partial R_f} &= \epsilon_f, \left| \frac{\partial f_4}{\partial R_f} \right| \leq K < \infty,\end{aligned}$$

Similarly ,differentiating f_5 we obtain:

$$\begin{aligned}\frac{\partial f_5}{\partial S_m} &= 0, \left| \frac{\partial f_5}{\partial S_m} \right| \leq K < \infty, & \frac{\partial f_5}{\partial S_f} &= 0, \left| \frac{\partial f_5}{\partial S_f} \right| \leq K < \infty, \\ \frac{\partial f_5}{\partial I_m} &= \gamma_m, \left| \frac{\partial f_5}{\partial I_m} \right| \leq K < \infty, & \frac{\partial f_5}{\partial I_f} &= 0, \left| \frac{\partial f_5}{\partial I_f} \right| \leq K < \infty, \\ \frac{\partial f_5}{\partial R_m} &= -m_1, \left| \frac{\partial f_5}{\partial R_m} \right| \leq K < \infty, & \frac{\partial f_5}{\partial R_f} &= 0, \left| \frac{\partial f_5}{\partial R_f} \right| \leq K < \infty,\end{aligned}$$

Lastly,if we Differentiate the last equation f_6 we obtain:

$$\begin{aligned}\frac{\partial f_6}{\partial S_m} &= 0, \left| \frac{\partial f_6}{\partial S_m} \right| \leq K < \infty, & \frac{\partial f_6}{\partial S_f} &= 0, \left| \frac{\partial f_6}{\partial S_f} \right| \leq K < \infty, \\ \frac{\partial f_6}{\partial I_m} &= 0, \left| \frac{\partial f_6}{\partial I_m} \right| \leq K < \infty, & \frac{\partial f_6}{\partial I_f} &= \gamma_f, \left| \frac{\partial f_6}{\partial I_f} \right| \leq K < \infty, \\ \frac{\partial f_6}{\partial R_m} &= 0, \left| \frac{\partial f_6}{\partial R_m} \right| \leq K < \infty, & \frac{\partial f_6}{\partial R_f} &= m_2, \left| \frac{\partial f_6}{\partial R_f} \right| \leq K < \infty,\end{aligned}$$

In conclusion, our analysis confirms that all partial derivatives are bounded and continuous. This property is crucial because it satisfies the conditions of the existence and uniqueness theorem for differential equations. By ensuring that the partial derivatives do not exhibit any discontinuities or unbounded behavior, we establish that the solution to the model equations is not only attainable but also singular. Thus, we can conclude that our approach is mathematically sound and reliable for further applications.

4.4 Positivity and Boundedness of a solution

To analyze the dynamics of smoking, it is crucial to understand how the solution evolves over time t . For modeling human diseases, it is essential to demonstrate that the solutions remain positive and bounded over an extended period. Let $t_0 > 0$, then for all $t \in [0, t_0]$, the values of the parameters can be chosen so that the solution $(S_m(t), S_f(t), I_m(t), I_f(t), R_m(t), R_f(t))$ to ODE with the positive initial conditions as defined

Theorem 4.1

Let $\Omega = (S_m, S_f, I_m, I_f, R_m, R_f) \mathbb{R}^7_+ : S_{m0} > 0, S_{f0} > 0, I_{m0} > 0, I_{f0} > 0, R_{m0} > 0, R_{f0} > 0$. Then the solutions of $(S_m(t), S_f(t), I_m(t), I_f(t), R_m(t), R_f(t))$ are positive for $t \geq 0$

Lemma 1: Grownwall Inequality

Let y and g be non-negative integrable functions and c a non negative constant. If

$$y(t) \leq c \int_0^t g(s)y(s)ds, \quad \text{for } t \geq 0$$

$$y(t) \leq c \exp\left(\int_0^t g(s)ds\right), \quad \text{for } t \geq 0$$

,then

$$y(t) \leq c \exp\left(\int_0^t g(s)ds\right), \quad \text{for } t \geq 0$$

Proof

Considering the system of differential equations, we will analyse the first equation (1) using the Grönwall Inequality.

$$\frac{dS_m}{dt} = \delta_m - \beta_m S_m I_m - \beta_f S_m I_f - \mu_m S_m$$

$$\frac{dS_m}{dt} \geq -\beta_m S_m I_m - \beta_f S_m I_f \Rightarrow$$

$$\int \frac{dS_m}{S_m} dS_m \geq \int -\beta_m I_m - \beta_f I_f dt$$

Upon applying separation of variables and including the initial condition, the solution is derived as follows:

$$S_m(t) \geq S_{m0} e^{\int (-\beta_m I_m - \beta_f I_f) dt} \geq 0$$

Similarly, we considered the equation (2), which is:

$$\frac{dS_f}{dt} = \delta_f - \beta_m S_f I_m - \beta_f S_f I_f - \mu_f S_f$$

$$\frac{dS_f}{dt} \geq -\beta_f S_f I_f - \beta_m S_f I_m \Rightarrow$$

$$\int \frac{dS_f}{S_f} \geq \int (-\beta_f I_f - \beta_m I_m) dt$$

Upon applying separation of variables and including the initial condition, the solution is derived as follows:

$$S_f(t) \geq S_{f0} e^{\int (-\beta_f I_f - \beta_m I_m) dt} \geq 0$$

Now, we consider the third equation (3), which is

$$\frac{dI_m}{dt} = \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m$$

,
It is true that

$$\frac{dI_m}{dt} \geq \beta_m S_m I_m - (\mu_m + \omega_m + \gamma_m) I_m \Rightarrow$$

$$\int \frac{dI_m}{I_m} \geq \int (\beta_m S_m - (\mu_m + \omega_m + \gamma_m)) dt$$

Upon applying separation of variables and including the initial condition, the solution is derived as follows:

$$I_m(t) \geq I_{m0} e^{\int \beta_m S_m - (\mu_m + \omega_m + \gamma_m) dt} \geq 0$$

Considering equation (4), which is:

$$\frac{dI_f}{dt} = \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f$$

It is true that:

$$\frac{dI_f}{dt} \geq \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f \Rightarrow$$

$$\int \frac{dI_f}{I_f} \geq \int (\beta_f S_f - (\mu_f + \omega_f + \gamma_f)) dt$$

Upon applying separation of variables and including the initial condition, the solution is derived as follows:

$$I_f(t) \geq I_{f0} e^{\int \beta_f S_f - (\mu_f + \omega_f + \gamma_f) dt} \geq 0$$

If we consider equation (5), which is:

$$\frac{dR_m}{dt} = \gamma_m I_m - (\mu_m + \omega_m + \epsilon_m) R_m$$

It is true that:

$$\frac{dR_m}{dt} \geq (\mu_m + \omega_m + \epsilon_m) R_m \Rightarrow$$

$$\int \frac{dR_m}{R_m} \geq \int (\mu_m + \omega_m + \epsilon_m) dt$$

Now employing separation of variables and incorporating the initial condition, the solution is obtained as follows:

$$R_m(t) \geq R_{m0} e^{\int (\mu_m + \omega_m + \epsilon_m) dt} \geq 0$$

Now if we consider equation (6) , which is:

$$\frac{dR_f}{dt} = \gamma_f I_f - (\mu_f + \omega_f + \epsilon_f) R_f$$

It is also true that:

$$\frac{dR_f}{dt} \geq (\mu_f + \omega_f + \epsilon_f) R_f \Rightarrow$$

$$\int \frac{dR_f}{R_f} \geq \int (\mu_f + \omega_f + \epsilon_f) dt$$

Upon applying separation of variables and including the initial condition, the solution is derived as follows:

$$R_f(t) \geq R_{f0} e^{\int (\mu_f + \omega_f + \epsilon_f) dt} \geq 0$$

This concludes the proof of the theorem. As a result, we can affirm that the solution of the model is non-negative and it is significant because it ensures that the values represented by the model, such as populations remain physically meaningful over time. This characteristic is essential for the validity of the model in real-world applications, as it reflects realistic scenarios where negative quantities are not applicable. Therefore, we can confidently state that the model maintains its integrity and relevance in describing the dynamics it aims to capture.

4.5 Smoking Free Equilibrium point(E_0)

The process of identifying the Smoking-Free Equilibrium(SFE) involves setting the right-hand side of the model equations to zero. This was done by assessing it when $I_m = I_f = R_m = R_f = 0$ and solving for the non-smoker variables. Consequently, we arrived at the Smoking-free equilibrium denoted as

$$E_0 = (S_m, S_f, I_m, I_f, R_m, R_f) = \left(\frac{\sigma_m}{\mu_m}, \frac{\sigma_f}{\mu_f}, 0, 0, 0, 0 \right)$$

4.6 Reproductive Number R_0

In this section, we explored the concept of the basic reproductive number, R_0 . We derived this essential parameter that governs the spread of smoking. To calculate R_0 , we employed the next-generation matrix method, which involves determining the spectral radius of the next-generation matrix. The model equations are reformulated, beginning with the newly infected classes.

Lemma

If x_0 represents a Smoking-Free Equilibrium (SFE) of equation and $F_i(x)$ satisfies the assumptions (H1) through (H5), then the derivatives $DF(x_0)$ and $DV(x_0)$ can be partitioned as:

$$DF(x_0) = \begin{bmatrix} F & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{and } DV(x_0) = \begin{bmatrix} V & 0 \\ J_3 & J_4 \end{bmatrix},$$

where F and V are $m \times m$ matrices defined by $F = \left[\frac{\partial F_i}{\partial x_j}(x_0) \right]$ and $V = \left[\frac{\partial V_i}{\partial x_j}(x_0) \right]$, Where $1 \leq i, j \leq m$. Additionally, F is non-negative, V is a non-singular M-matrix, and all eigenvalues of J_4 have positive real parts. Consequently, the matrix V^{-1} is non-negative, as is FV^{-1} . If an infected individual is introduced into compartment k of a disease-free population, then the (j, k) entry of V^{-1} can be interpreted as the average length of time an individual spends in compartment j during its life time. The (i, j) entry of F can be interpreted as the rate at which infected individuals in compartment j produce new infections in compartment i .

The matrix FV^{-1} is referred to as the next-generation matrix for the model. The (i, k) entry of the next-generation matrix represents the expected number of new infections in compartment i produced by the infected individual originally placed into compartment k . The basic reproduction number, R_0 , is calculated as:

$$R_0 = (FV^{-1}),$$

where (FV^{-1}) denotes the spectral radius of FV^{-1} . R_0 serves as a threshold parameter for the stability of the Smoking free Equilibrium SFE

Proof

From our system of equations

$$\frac{dI_m}{dt} = \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m \quad (3)$$

$$\frac{dI_f}{dt} = \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f \quad (4)$$

Now, by the principle of the next-generation matrix, we obtained:

$$f = \begin{bmatrix} \beta_m S_m I_m + \beta_f S_m I_f \\ \beta_m S_f I_m + \beta_f S_f I_f \end{bmatrix}$$

and

$$v = \begin{bmatrix} (\mu_m + \omega_m + \gamma_m)I_m + \epsilon_m R_m \\ (\mu_f + \omega_f + \gamma_f)I_f + \epsilon_f R_f \end{bmatrix}$$

The matrices of f and v evaluated at Smoking free are given by F and V , respectively, such that:

$$F = \begin{pmatrix} \frac{\beta_m \sigma_m}{\mu_m} & \frac{\beta_f \sigma_m}{\mu_f} \\ \frac{\beta_m \sigma_f}{\mu_f} & \frac{\beta_f \sigma_f}{\mu_f} \end{pmatrix}$$

, and

$$V = \begin{pmatrix} \mu_m + \omega_m + \gamma_m & 0 \\ 0 & \mu_f + \omega_f + \gamma_f \end{pmatrix}$$

Now the inverse of V is given by:

$$V^{-1} = \begin{pmatrix} \frac{1}{\mu_m + \omega_m + \gamma_m} & 0 \\ 0 & \frac{1}{\mu_f + \omega_f + \gamma_f} \end{pmatrix}$$

Now we multiply F and v^{-1} and we get:

$$FV^{-1} = \begin{pmatrix} \frac{\beta_m \sigma_m}{\mu_m(\mu_m + \omega_m + \gamma_m)} & \frac{\beta_f \sigma_m}{\mu_m(\mu_f + \omega_f + \gamma_f)} \\ \frac{\beta_m \sigma_f}{\mu_f(\mu_m + \omega_m + \gamma_m)} & \frac{\beta_f \sigma_f}{\mu_f(\mu_f + \omega_f + \gamma_f)} \end{pmatrix}$$

The eigenvalues of FV^{-1} are :

$$\lambda_1 = 0 \quad \text{and} \quad \lambda_2 = \frac{\beta_m \sigma_m}{\mu_m(\mu_m + \omega_m + \gamma_m)} + \frac{\beta_f \sigma_f}{\mu_f(\mu_f + \omega_f + \gamma_f)}$$

The primary eigenvalues of FV^{-1} correspond to the fundamental reproduction rates of smoking. These values indicate the typical number of new infections originating from each infected person, Therefore, the fundamental reproduction rate of smoking is:

$$R_0 = \frac{\beta_m \sigma_m}{\mu_m(\mu_m + \omega_m + \gamma_m)} + \frac{\beta_f \sigma_f}{\mu_f(\mu_f + \omega_f + \gamma_f)}$$

can be written as

$$R_0 = R_{0m} + R_{0f}$$

4.7 Local Stability of Smoking Free Equilibrium

We establish E_0 , which represents the point when there are no smokers. At this point, the system has a constant solution $E_0 = (\frac{\sigma_m}{\mu_m}, \frac{\sigma_f}{\mu_f}, 0, 0, 0, 0)$.

proposition 1

The smoking-free equilibrium point is locally asymptotically stable if $R_0 < 1$ and unstable if $R_0 > 1$.

Proof

To prove this theorem, first, we obtain the Jacobian matrix of the system at the smoking-free equilibrium E_0 as follows:

$$\begin{pmatrix} -\mu_m & 0 & -k_1 & -k_2 & 0 & 0 \\ 0 & -\mu_f & -k_3 & -k_4 & 0 & 0 \\ 0 & 0 & k_1 - m_1 & k_2 & \epsilon_m & 0 \\ 0 & 0 & k_3 & k_4 - m_2 & 0 & \epsilon_f \\ 0 & 0 & \gamma_m & 0 & -m_1 & 0 \\ 0 & 0 & 0 & \gamma_f & 0 & -m_2 \end{pmatrix}$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$, $k_1 = \frac{\beta_m \sigma_m}{\mu_m}$, $k_2 = \frac{\beta_f \sigma_m}{\mu_m}$, $k_3 = \frac{\beta_m \sigma_f}{\mu_f}$, $k_4 = \frac{\beta_f \sigma_f}{\mu_f}$

We derived a characteristic polynomial from the Jacobian matrix of equation

$(\mu_m - \lambda)(\mu_f - \lambda)(a\lambda^4 + b\lambda^3 + c\lambda^2 + d\lambda + g) = 0$, where
 $b = m_1(2 - R_{0m}) + m_2(2 - R_{0f})$

$$c = k_3 k_2 + m_1 m_2 + [k_1(\frac{1}{R_{0m}} - 1) + k_4 + \frac{k_4}{R_{0f}}](\frac{k_1}{R_{0m}} + \frac{k_4}{R_{0f}}) + k_1(\frac{1}{R_{0m}} - 1)(k_4 + \frac{k_4}{R_{0f}})$$

$$d = \epsilon_m \gamma_m k_4(\frac{1}{R_{0f}} - 1) + k_2 k_3(\frac{k_1}{R_{0m}}) + \epsilon_f \gamma_f k_1(\frac{1}{R_{0m}} - 1) + \epsilon_f \gamma(\frac{1}{R_{0m}} - 1) + k_2 K_3 k_4(\frac{1}{R_{0f}}) + k_1^2 K_4(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0f}} - 1)(\frac{1}{R_{0m}}) + 2k_1 k_4^2(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0f}} - 1)(\frac{1}{R_{0f}}) + \epsilon_m \gamma_m k_4(\frac{1}{R_{0f}}) + k_1^2 k_4(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0m}})(\frac{1}{R_{0f}}) + k_1 k_4^2(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0f}})(\frac{1}{R_{0f}})$$

$$g = \epsilon_m \epsilon_f \gamma_m \gamma_f + m_1 m_2 (k_1 - m_1)(k_4 - m_2) + m_1 \epsilon_f (k_1 - m_1) + \gamma_f + m_2 \epsilon_m \gamma_m (k_4 - m_2) - m_1 m_2 k_2 k_3$$

From the characteristic equation, we can see that

$$-\mu_m - \lambda = 0, \mu_f - \lambda = 0, (a\lambda^4 + b\lambda^3 + c\lambda^2 + d\lambda + g) = 0$$

Now we observe that

$$\lambda = -\mu_m < 0 \quad \lambda = -\mu_f < 0$$

and for the the expression $(a\lambda^4 + b\lambda^3 + c\lambda^2 + d\lambda + g) = 0$

We applied Routh-Hurwitz criteria. By the principle of Routh-Hurwitz criteria, The expression has strictly negative real roots if and only if $a > 0, c > 0, d > 0$, and $abc > c^2 + ad$.

Clearly we can see that $b = m_1(2 - R_{0m}) + m_2(2 - R_{0f})$ is positive if $R_{0m} < 1$ and $R_{0f} < 1$. Also $c = k_3 k_2 + m_1 m_2 + [k_1(\frac{1}{R_{0m}} - 1) + k_4 + \frac{k_4}{R_{0f}}](\frac{k_1}{R_{0m}} + \frac{k_4}{R_{0f}}) + k_1(\frac{1}{R_{0m}} - 1)(k_4 + \frac{k_4}{R_{0f}})$ and $d = \epsilon_m \gamma_m k_4(\frac{1}{R_{0f}} - 1) + k_2 k_3(\frac{k_1}{R_{0m}}) + \epsilon_f \gamma_f k_1(\frac{1}{R_{0m}} - 1) + \epsilon_f \gamma(\frac{1}{R_{0m}} - 1) + k_2 K_3 k_4(\frac{1}{R_{0f}}) + k_1^2 K_4(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0f}} - 1)(\frac{1}{R_{0m}}) + 2k_1 k_4^2(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0f}} - 1)(\frac{1}{R_{0f}}) + \epsilon_m \gamma_m k_4(\frac{1}{R_{0f}}) + k_1^2 k_4(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0m}})(\frac{1}{R_{0f}}) + k_1 k_4^2(\frac{1}{R_{0m}} - 1)(\frac{1}{R_{0m}})(\frac{1}{R_{0f}})$ are positive if $R_{0m} < 1$ and $R_{0f} < 1$. Therefore the smoking free equilibrium will be locally asymptotically stable if and only if $R_0 < 1$

4.8 Global stability Analysis for E_0

The Smoking free equilibrium point (E_0) for the system is globally asymptotically stable provided $R_0 < 1$ and unstable if $R_0 > 1$

Proof

Following [14], we write the system in the form:

$$X'(t) = F(X, Y),$$

$$Y'(t) = G(X, Y), \quad G(X, 0) = 0$$

where $X = (S_m, S_f, R_m, R_f)$ and $Y = (I_m, I_f)$. Here the components of $X \in \mathbb{R}_+^4$ denote the number of non-smokers and the components of $Y \in \mathbb{R}_+^2$ denote the number of smokers. The smoking-free equilibrium is now denoted by E_0 . We have to prove two conditions,

(K1) For $X(t) = F(X, 0)$, X is globally asymptotically stable;

(K2) $G(X, Y) = UYG(X, Y)$, $\hat{G}(X, Y), \hat{G}(X, Y) \geq 0$ for $(X, Y) \in \Phi$ are satisfied, where Φ_1 is a positively invariant attracting domain.

Theorem

The fixed point $U_0 = (x^*, 0)$ is a globally asymptotically stable equilibrium of provided that $R_0 < 1$ and assumptions (K1) and (K2) are satisfied.

Proof

$$F(X, Y) = \begin{bmatrix} \delta_m - \beta_m S_m I_m - \beta_f S_m I_f - \mu_m S_m \\ \delta_f - \beta_m S_f I_m - \beta_f S_f I_f - \mu_f S_f \\ \gamma_m I_m - (\mu_m + \omega_m + \epsilon_m) R_m \\ \gamma_f I_f - (\mu_f + \omega_f + \epsilon_f) R_f \end{bmatrix}$$

Now we substitute the smoking free equilibrium point(E_0) in $F(X, Y)$ we have,

$$F(X, 0) = \begin{bmatrix} \delta_m - \mu_m S_m \\ \delta_f - \mu_f S_f \\ 0 \\ 0 \end{bmatrix} \geq 0$$

Since $S_m(t) \leq \frac{\sigma_m}{\mu_m}$ and $S_f(t) \leq \frac{\sigma_f}{\mu_f}$ it follows that $F(X, 0) \geq 0$ thus the condition (K1) is satisfied.

Now we find U

$$\hat{G}(X, Y) = \begin{bmatrix} \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m) I_m + \epsilon_m R_m \\ \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f) I_f + \epsilon_f R_f \end{bmatrix}$$

Finding the Jacobian of U and substitute the smoking free equilibrium point we have,

$$U = \begin{bmatrix} \frac{\beta_m \sigma_m}{\mu_m} - (\mu_m + \omega_m + \gamma_m) & \frac{\beta_f \sigma_m}{\mu_f} \\ \frac{\beta_f \sigma_f}{\mu_f} - (\mu_f + \omega_f + \gamma_f) & \frac{\beta_m \sigma_f}{\mu_f} \end{bmatrix}$$

Now we find UY as follows:

$$UY = \begin{bmatrix} \frac{\beta_m \sigma_m}{\mu_m} - (\mu_m + \omega_m + \gamma_m) & \frac{\beta_f \sigma_m}{\mu_f} \\ \frac{\beta_f \sigma_f}{\mu_f} - (\mu_f + \omega_f + \gamma_f) & \frac{\beta_m \sigma_f}{\mu_f} \end{bmatrix} \begin{bmatrix} I_m \\ I_f \end{bmatrix}$$

This simplifies to

$$UY = \begin{bmatrix} [\frac{\beta_m \sigma_m}{\mu_m} - (\mu_m + \omega_m + \gamma_m)]I_m + [\frac{\beta_f \sigma_m}{\mu_m}]I_f \\ [\frac{\beta_f \sigma_f}{\mu_f} - (\mu_f + \omega_f + \gamma_f)]I_m + [\frac{\beta_m \sigma_f}{\mu_f}]I_f \end{bmatrix}$$

Now we use UY and $\hat{G}(X, Y)$ to find $G(X, Y)$

$$G(X, Y) = UY\hat{G}(X, Y)$$

$$G(X, Y) = \begin{bmatrix} [\frac{\beta_m \sigma_m}{\mu_m} - (\mu_m + \omega_m + \gamma_m)]I_m + [\frac{\beta_f \sigma_m}{\mu_m}]I_f \\ [\frac{\beta_f \sigma_f}{\mu_f} - (\mu_f + \omega_f + \gamma_f)]I_m + [\frac{\beta_m \sigma_f}{\mu_f}]I_f \end{bmatrix} - \begin{bmatrix} \beta_m S_m I_m + \beta_f S_m I_f - (\mu_m + \omega_m + \gamma_m)I_m + \epsilon_m R_m \\ \beta_m S_f I_m + \beta_f S_f I_f - (\mu_f + \omega_f + \gamma_f)I_f + \epsilon_f R_f \end{bmatrix}$$

$$G(X, Y) = \begin{bmatrix} \beta_m \sigma_m I_m + \beta_f \sigma_m I_f - \beta_m S_m I_m + \beta_f S_m I_f + \epsilon_m R_m \\ \beta_f I_m \sigma_f + \beta_m \sigma_f I_f - \beta_f S_f I_f - \beta_m S_f I_m - (I_m + I_f)(\mu_f + \omega_f + \gamma_f) - \epsilon_f R_f \end{bmatrix} \geq 0$$

As highlighted earlier, $S_m(t) \leq \frac{\sigma_m}{\mu_m}$ and $S_f(t) \leq \frac{\sigma_f}{\mu_f}$ for all $t \geq 0$, thus condition(K2) is satisfied. Therefore, one can conclude that the smoking-free equilibrium point E_0 is globally asymptotically stable.

4.9 Endemic equilibrium points

The endemic equilibrium points are given by

$$(E_1) = (S_m^*, S_f^*, I_m^*, I_f^*, R_m^*, R_f^*) = (\frac{\sigma_m}{\beta_m I_m^* + \beta_f S_f^* + \mu_m}, \frac{\sigma_f}{\beta_f I_f^* + \beta_m S_m^* + \mu_f}, \frac{\beta_f S_m^* I_f^* + \epsilon_m R_m^*}{\beta_m S_m^* - m_1}, \frac{\beta_m S_f^* I_m^* + \epsilon_f R_f^*}{\beta_f S_f^* - m_2}, \frac{\gamma_m I_m^*}{m_1}, \frac{\gamma_f I_f^*}{m_2})$$

4.9.1 Local Stability of the Endemic Equilibrium

Theorem

. The endemic equilibrium, E_1 , of system Equation is locally asymptotically stable whenever $R_0 > 1$, and unstable if $R_0 < 1$.

Proof

. The following variational matrix is computed at E_1 to determine its local stability. That is

$$J(E_1) = \begin{bmatrix} b_{11} & 0 & b_{13} & b_{14} & 0 & 0 \\ 0 & b_{22} & b_{23} & b_{24} & 0 & 0 \\ b_{31} & 0 & b_{33} & b_{34} & b_{35} & 0 \\ 0 & b_{42} & 0 & b_{44} & 0 & b_{46} \\ 0 & 0 & b_{53} & 0 & b_{55} & 0 \\ 0 & 0 & 0 & b_{64} & 0 & b_{66} \end{bmatrix}$$

where,

$$\begin{aligned} b_{11} &= -\beta_m I_m^* - \beta_f I_f^* - \mu_m & b_{13} &= -\beta_m S_m^* & b_{14} &= \beta_f S_m^* & b_{22} &= -\beta_m I_m^* - \beta_f I_f^* - \mu_f & b_{23} &= -\beta_m S_f^* \\ b_{24} &= -\beta_f S_f^* & b_{31} &= \beta_m I_m^* + \beta_f I_f^* & b_{33} &= \beta_m S_m^* - m_1 & b_{34} &= \beta_f S_m^* & b_{35} &= \epsilon_m & b_{42} &= \beta_f I_f^* + \\ & \beta_m S_f^* & b_{44} &= \beta_f S_f^* - m_2 & b_{46} &= \epsilon_f & b_{53} &= \gamma_m & b_{55} &= -m_1 & b_{64} &= \gamma_f & b_{66} &= -m_2 \end{aligned}$$

The characteristic equation corresponding to this variational matrix is given by the following polynomial equation:

$$\lambda^6 + a_1\lambda^5 + a_2\lambda^4 + a_3\lambda^3 + a_4\lambda^2 + a_5\lambda + a_6$$

, where

$$\begin{aligned} a_1 &= b_{11} + b_{22} + b_{44} + b_{55} + b_{66} + b_{33} \\ a_2 &= -b_{46}b_{64} - b_{35}b_{53} - b_{11}b_{55} - b_{22} - b_{33}b_{55} - b_{44}b_{55} - b_{11}b_{22} + b_{13}b_{31} - b_{11}b_{33} - b_{22}b_{33} - b_{11}b_{44} - b_{22}b_{44} - b_{33}b_{44} \\ a_3 &= b_{11}b_{35}b_{53} + b_{22}b_{35}b_{53} + b_{35}b_{44}b_{53} + b_{11}b_{46}b_{64} + b_{22} + b_{46}b_{64} + b_{33}b_{46}b_{64} + b_{46}b_{55}b_{64} + b_{35}b_{53}b_{66} + b_{11}b_{22}b_{55} - \\ & b_{13}b_{31}b_{55} + b_{11}b_{33}b_{55} + b_{22}b_{33}b_{55} + b_{11}b_{44}b_{55} + b_{22}b_{44}b_{55} + b_{33}b_{44}b_{55} + b_{13}b_{22}b_{31} + b_{11}b_{22}b_{33} + b_{11}b_{22}b_{44} - \\ & b_{13}b_{31}b_{44} + b_{11}b_{33}b_{44} + b_{22}b_{33}b_{44} + b_{11}b_{24}b_{42} + b_{24}b_{33}b_{42} - b_{23}b_{34}b_{42} \\ a_4 &= b_{11}b_{22}b_{46}b_{64} + b_{13}b_{31}b_{46}b_{64} - b_{11}b_{33}b_{46}b_{64} - b_{22}b_{33}b_{46}b_{64} + b_{35}b_{46}b_{53}b_{64} - b_{11}b_{46}b_{55}b_{64} - b_{22}b_{46}b_{55}b_{64} - \\ & b_{33}b_{46}b_{55}b_{64} - b_{11}b_{35}b_{53}b_{66} - b_{22}b_{35}b_{53}b_{66} - b_{35}b_{44}b_{53}b_{66} - b_{11}b_{22}b_{35}b_{53} + b_{24}b_{35}b_{42}b_{53} - b_{11}b_{35}b_{44}b_{53} - b_{22}b_{35}b_{44}b_{53} + \\ & b_{13}b_{22}b_{31}b_{44} - b_{11}b_{22}b_{33}b_{44} + b_{13}b_{22}b_{31}b_{55} - b_{11}b_{22}b_{33}b_{55} - b_{11}b_{22}b_{44}b_{55} + b_{13}b_{31}b_{44}b_{55} - b_{11}b_{33}b_{44}b_{55} - b_{22}b_{33}b_{44}b_{55} \\ a_5 &= b_{13}b_{22}b_{31}b_{46}b_{64} + b_{11}b_{22}b_{33}b_{46}b_{64} - b_{11}b_{35}b_{46}b_{53}b_{64} - b_{22}b_{35}b_{46}b_{53}b_{64} + b_{11}b_{22}b_{46}b_{55}b_{64} - b_{13}b_{31}b_{46}b_{55}b_{64} + \\ & b_{11}b_{33}b_{46}b_{55}b_{64} + b_{22}b_{33}b_{46}b_{55}b_{64} + b_{11}b_{22}b_{35}b_{53}b_{66} - b_{24}b_{35}b_{42}b_{53}b_{66} + b_{11}b_{35}b_{44}b_{53}b_{66} + b_{22}b_{35}b_{44}b_{53}b_{66} - \\ & b_{11}b_{24}b_{35}b_{42}b_{53} + b_{11}b_{22}b_{35}b_{44}b_{53} \\ a_6 &= -b_{14}b_{23}b_{31}b_{42} + b_{13}b_{24}b_{31}b_{42} - b_{11}b_{24}b_{33}b_{42} + b_{11}b_{23}b_{34}b_{42} - b_{13}b_{22}b_{31}b_{44}b_{55} + b_{11}b_{22}b_{33}b_{44}b_{55} + \\ & b_{11}b_{22}b_{35}b_{46}b_{53}b_{64} + b_{13}b_{22}b_{31}b_{46}b_{55}b_{64} - b_{11}b_{22}b_{33}b_{46}b_{55}b_{64} + b_{11}b_{24}b_{35}b_{42}b_{53}b_{66} - b_{11}b_{22}b_{35}b_{44}b_{53}b_{66} \end{aligned}$$

By using Routh-Hurwitz criteria the roots of the above polynomial equation will have negative real parts if

$$\begin{aligned} \begin{bmatrix} 1 & a_2 \\ a_1 & a_3 \end{bmatrix} &> 0, \quad \begin{bmatrix} 1 & a_2 & a_4 \\ a_1 & a_3 & a_5 \\ 0 & 1 & a_2 \end{bmatrix} > 0 \\ \begin{bmatrix} 1 & a_2 & a_4 & a_6 \\ a_1 & a_3 & a_5 & 0 \\ 0 & 1 & a_2 & a_4 \\ 0 & a_1 & a_3 & a_5 \end{bmatrix} &> 0, \quad \begin{bmatrix} 1 & a_2 & a_4 & a_6 & 0 \\ a_1 & a_3 & a_5 & 0 & 0 \\ 0 & 1 & a_2 & a_4 & a_6 \\ 0 & a_1 & a_3 & a_5 & 0 \\ 0 & 0 & 1 & a_2 & a_4 \end{bmatrix} > 0 \\ \begin{bmatrix} 1 & a_2 & a_4 & a_6 & 0 & 0 \\ a_1 & a_3 & a_5 & 0 & 0 & 0 \\ 0 & 1 & a_2 & a_4 & a_6 & 0 \\ 0 & a_1 & a_3 & a_5 & 0 & 0 \\ 0 & 0 & 1 & a_2 & a_4 & 0 \\ 0 & 0 & a_1 & a_3 & a_5 & a_6 \end{bmatrix} &> 0 \end{aligned}$$

Hence, the equilibrium point E_1 is locally asymptotically stable if all the above conditions are satisfied.

4.10 Sensitivity analysis of R_0

The sensitivity analysis is employed to assess the importance of individual parameters in the dynamics of smoking model. Understanding the relative significance of different factors linked to the spread of smoking is essential for effectively mitigating human mortality and morbidity resulting from their spread. This research utilizes numerical sensitivity indices top in point parameters that notably influence the basic reproduction number, R_0 , and therefore warrant attention in intervention planning. The

sensitivity analysis methodology in this thesis aligns with the approach initially introduced by. [24].

Definition

The normalized forward sensitivity index of a function u that depends differentiably on a parameter p is defined as:

$$\Gamma_u^p = \frac{\partial u}{\partial p} \cdot \frac{p}{u}$$

. Considering that R_0 has an explicit formula, we calculate the sensitivity of R_0 as

$$\Gamma_{R_0}^p = \frac{\partial R_0}{\partial p} * \frac{p}{R_0}$$

.

Sensitivity of R_0

For example, the sensitivity index of R_0 with respect to μ_m is:

$$\Gamma_{R_0}^{\mu_m} = \frac{\partial R_0}{\partial \mu_m} * \frac{\mu_m}{R_0} = -\frac{\beta_m \sigma_m \mu_f m_2 (2\mu_m + \omega_m + \gamma_m)}{m_1 (\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} < 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to μ_f is:

$$\Gamma_{R_0}^{\mu_f} = \frac{\partial R_0}{\partial \mu_f} * \frac{\mu_f}{R_0} = -\frac{\beta_f \sigma_f \mu_m m_1 (2\mu_f + \omega_f + \gamma_f)}{m_2 (\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} < 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to ω_m is:

$$\Gamma_{R_0}^{\omega_m} = \frac{\partial R_0}{\partial \omega_m} * \frac{\omega_m}{R_0} = -\frac{\beta_m \sigma_m \omega_m \mu_f m_2}{m_1 (\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} < 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to ω_f is:

$$\Gamma_{R_0}^{\omega_f} = \frac{\partial R_0}{\partial \omega_f} * \frac{\omega_f}{R_0} = -\frac{\beta_f \sigma_f \omega_f \mu_m m_1}{m_2 (\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} < 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to γ_m is:

$$\Gamma_{R_0}^{\gamma_m} = \frac{\partial R_0}{\partial \gamma_m} * \frac{\gamma_m}{R_0} = -\frac{\beta_m \sigma_m \gamma_m \mu_f m_2}{m_1 (\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} < 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to γ_f is:

$$\Gamma_{R_0}^{\gamma_f} = \frac{\partial R_0}{\partial \gamma_f} * \frac{\gamma_f}{R_0} = -\frac{\beta_f \sigma_f \gamma_f \mu_m m_1}{m_2 (\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} < 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to β_m is:

$$\Gamma_{R_0}^{\beta_m} = \frac{\partial R_0}{\partial \beta_m} * \frac{\beta_m}{R_0} = \frac{\beta_m \sigma_m}{(\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} > 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to β_f is:

$$\Gamma_{R_0}^{\beta_f} = \frac{\partial R_0}{\partial \beta_f} * \frac{\beta_f}{R_0} = \frac{\beta_f \sigma_f \mu_m m_1}{(\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} > 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to σ_m is:

$$\Gamma_{R_0}^{\sigma_m} = \frac{\partial R_0}{\partial \sigma_m} * \frac{\sigma_m}{R_0} = \frac{\beta_m \sigma_m \mu_f m_2}{(\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} > 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

the sensitivity index of R_0 with respect to ω_f is:

$$\Gamma_{R_0}^{\sigma_f} = \frac{\partial R_0}{\partial \sigma_f} * \frac{\sigma_f}{R_0} = \frac{\beta_f \sigma_f \mu_m m_1}{(\mu_m \beta_m \sigma_m m_2 + \mu_f \beta_f \sigma_f m_1)} > 0$$

where $m_1 = (\mu_m + \omega_m + \gamma_m)$, $m_2 = (\mu_f + \omega_f + \gamma_f)$

4.10.1 sensitivity indices table

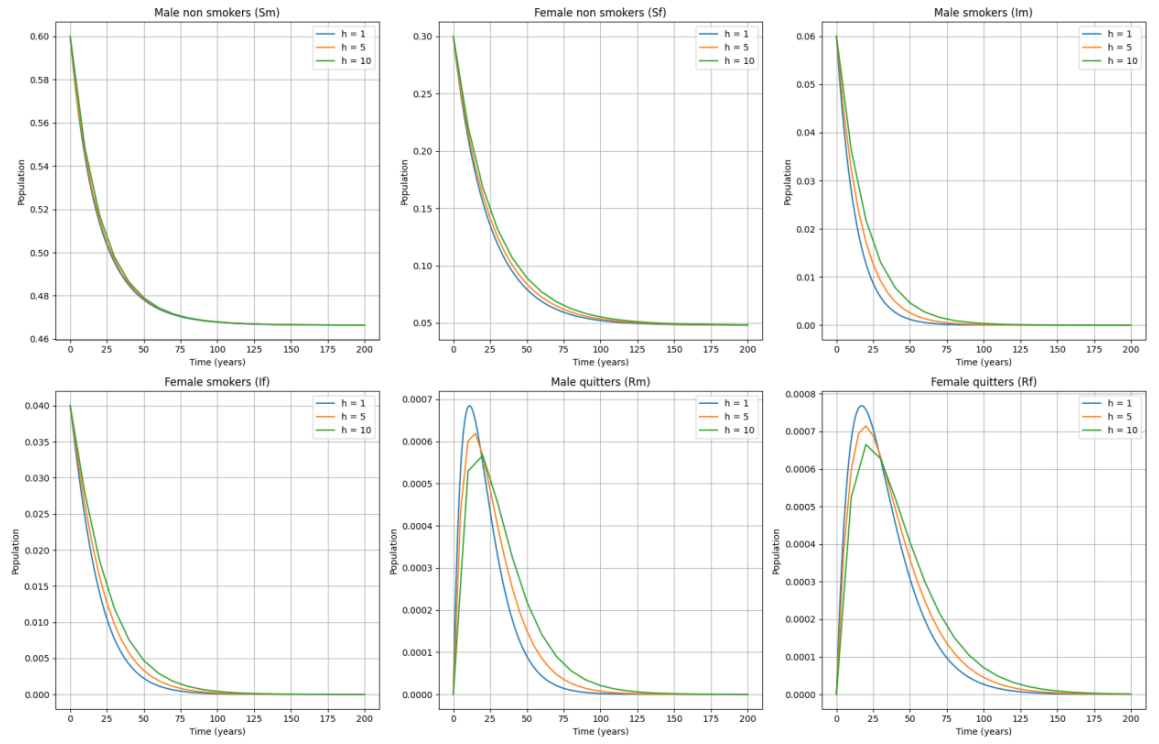
Parameter	Sensitivity index
μ_m	-1,36
μ_f	-0.00232
ω_m	-0.63
ω_f	-0.00071
γ_m	-0.025
γ_f	-0.000058
β_m	+1
β_f	+0.00155
σ_m	+1
σ_f	+0.00155

From the sensitivity analysis we observe that a 1% increase in μ_m and μ_f will reduce R_0 by 1.36% and 0.00232% respectively which means as new deaths occur, the population size is reduced. Similarly a 1% increase in ω_m and ω_f will reduce R_0 by 0.63% and 0.00071% respectively which means as new deaths due to smoking related diseases occur, the number of people to pass smoking is reduced. If we increase γ_m and γ_f by 1%, R_0 is reduced by 0.025% and 0.0000058% respectively which implies that as individuals quit smoking, the number of new infections is also reduced. Moreover, a 1% increase in β_m and β_f will increase the reproductive number by 1% and 0.00155% respectively which means that an increase in the rate of transmission will increase the number of new smokers. Similarly, a 1% increase in σ_m and σ_f will increase the reproductive number by 1% and 0.00155% respectively. We also noticed that males contribute much to R_0 and females have a smaller contribution.

4.11 Numerical simulations

The mathematical analysis of this dynamics of smoking model has been conducted to examine the effects of various parameters in this smoking model (equations 1-6), several numerical simulations were performed, focusing on scenarios where $R_0 < 1$. The diagrams below illustrate the convergence of the solution towards the disease-free equilibrium using the Nonstandard Finite Difference (NSFD) scheme with step sizes $h = 1, h = 5$ and $h = 10$ where $\phi = \phi(h) + O(h^2)$. NSFD scheme was used because it provides better numerical stability for stiff problems, reducing the risk of numerical oscillations and they can ensure that the solution remains non-negative, which is essential in our model since it considers non-negative populations. In addition NSFD schemes tend to converge to the correct equilibrium points of the continuous system and it preserves the qualitative behavior of the model. NSFD can maintain certain properties of the original differential equations, such as conservation laws or invariant sets, which are important for accurate modeling. NSFD methods can often be implemented easily in computational algorithms, providing a straightforward approach to numerical simulation. Using Python as our main computational tool, we leverage its versatility in developing algorithms, visualizing data, conducting analyses, and performing numerical computations. This language creates an optimal environment for efficiently handling computationally intensive tasks. We use the following values of parameters to compute our simulations $\beta_m = 0.05$, $\beta_f = 0.0005$, $\gamma_m = 0.0031$, $\gamma_f = 0.0031$, $\mu_m = 0.04288$, $\mu_f = 0.0415744$, $\omega_m = 0.0642$, $\omega_f = 0.0228$, $\delta_m = 0.02$, $\delta_f = 0.002$, $\epsilon_m = 0.006$, $\epsilon_f = 0.0006$. The initial conditions $(S_{m0}, S_{f0}, I_{m0}, I_{f0}, R_{m0}, R_{f0}) = (0.6, 0.3, 0.06, 0.04, 0, 0)$. The following are the results of the simulations

Smoking Epidemic Model Dynamics with NSFD Scheme ($R_0 < 1$)



From the numerical simulations we discover that the potential smoker population S_m and S_f converges to the smoking free equilibrium points $S_m = \frac{\sigma_m}{\mu_m}$ and $S_f = \frac{\sigma_f}{\mu_f}$ respectively. The infected populations I_m and I_f converges to zero indicating that the smoking epidemic dies out with respect to time as the populations moves to the quitter compartment. Recovered populations R_m and R_f also approach zero. Male population dominates the epidemic due to higher transmission rate and recruitment rate. By decreasing the value of h we increase the convergence rate of the potential smoker populations and as h increases with time convergence is delayed. Similarly, if we decrease the value of h , the convergence of the infected population increases. If we decrease the value of h on the recovered population, the population increases and eventually decreases rapidly with time compared with a large h . It can be easily noticed that by reducing the step size the system converge rapidly to the steady state point. In conclusion, the number of these populations will decrease and reach their equilibrium position, which ensure that the proposed model is stable. The biological interpretation of these results shows that if $R_0 < 1$, the population is stable which agree with the results obtained from the qualitative analysis.

Chapter 5

Discussion and Conclusion

5.1 Key findings

From this research we discover that smoking have negative effects to both smokers and on smokers. The effects of smoking include economic effects, social effects and health effects. We also discover that smoking is dangerous and can cause various diseases such as lung cancer, Chronic Obstructive Pulmonary Disease (COPD) and Cardiovascular Diseases. In model formulation we discovered that there is smoking related mortality. We developed a mathematical model to examine smoking behaviors across different genders. The model examines the interactions between potential males and female smokers, male and female smokers, and male and female quitters, highlighting the factors influencing smoking prevalence. We discovered that smoking can spread from one individual to another through interaction. The interaction between female smokers and male non smokers can spread the smoking behaviour and vice versa. We also noticed that people who quit smoking have a chance of relapsing to smoking again thus people may quit temporarily. We found that the smoking-free equilibrium is locally asymptotically stable when $R_0 = \frac{\beta_m \sigma_m}{\mu_m(\mu_m + \omega_m + \gamma_m)} + \frac{\beta_f \sigma_f}{\mu_f(\mu_f + \omega_f + \gamma_f)}$ is below 1. In this case the number of new smokers will remain low therefore it is necessary to keep R_0 as small as possible and this indicates that, under certain conditions, it is possible to eradicate smoking from the population. We noticed that higher contact rates between smokers and non-smokers increase the likelihood of smoking initiation therefore there is need to manage the contact rate. To decrease the number of smokers, we can manage the contact rates β_m and β_f as well as the recruitment rate σ_m and σ_f . In this model higher recovery rate has a positive effect in reducing smoking therefore increasing γ_m and γ_f will reduce R_0 hence maintain stability. If R_0 is above 1 the model predicts an endemic equilibrium where smoking persists in the population, which means that the number of new smokers in the society will be high and it will be difficult to reduce smoking related diseases and deaths. In this research, it is discovered that there is mortality due to smoking related diseases. From sensitivity analysis, the model demonstrates that local asymptotic stability for the smoking-free equilibrium is sensitive to parameter variations, particularly in contact and recovery rates. We noticed that there is male dominance in the dynamics. The male contributed much in the spread of smoking due to their high recruitment rate (σ_m) and high rate at which individuals become smokers due to interaction with male smokers (β_m) and and most smokers in the society are males. Initial dynamics show rapid increases in smoking rates, followed by stabilization, highlighting the importance of early interventions. The model predicts that sustained interventions can significantly reduce smoking prevalence over time.

5.2 Limitations of the study

- Simplified Assumptions

The model rely on simplified assumptions about population dynamics, behaviors, and interac-

tions, which may not fully capture the complexity of real-world smoking behaviors. Assuming homogeneous mixing of population may ignore real-world social networks (e.g., family, workplace clusters) where smoking spreads non-randomly.

- Fixed parameters

The parameters used in this model are fixed which ignores the fact that these parameters may change due to change in policies, impacts of control measures and many other factors. Also, the accuracy of the model's predictions is highly dependent on the estimation of parameters. The research is constrained by the availability and quality of data used for modeling.

- No age structure

Our model overlooks the fact that people of different ages behave differently and it assumes a constant value of parameters among all individuals.

- Limited Scope of Factors

The model did not take into account for all relevant socio-economic, cultural, and environmental factors that influence smoking behavior, overlooking key determinants of smoking prevalence.

- Intervention strategies

The effectiveness of interventions are not considered in this model and they may vary across different demographic groups, and the model did not capture these variations comprehensively.

5.3 Recommendations

Based on the findings from our model, the following recommendations are proposed to effectively reduce smoking prevalence and improve public health outcomes.

1. Targeted Public Health Interventions

Increase Awareness Campaigns: Focus on educating the population about the risks of smoking and the benefits of cessation. **Implement Smoking Cessation Programs:** Provide accessible and effective support systems for individuals attempting to quit, including counseling and pharmacological aids.

2. Reduce Contact Rates β_m and β_f

The contact rate can be reduced by enforcing smoking restrictions through implementation and enforcement of policies that limit smoking in public places to reduce exposure of non-smokers. Implementing social norms and policies that restrict smoking in public places, along with targeted outreach, can mitigate the spread of smoking behavior. In addition, there is a need to encourage community initiatives that foster a culture of non-smoking among youths. Collaborating with local organizations to implement interventions that resonate with specific community values and needs is a necessity.

3. Enhance Recovery Rates

The recovery rates can be reduced if we ensure that resources for quitting smoking such as Varenicline, Nicotine replacement therapy, bupropion and Cognitive behavioral therapy are widely available in public health institutions. Policymakers need to develop programs to support individuals in maintaining their non-smoking status and prevent relapse and they must focus on providing equal access to cessation resources in every community. It is beneficial to utilize social media platforms to reach younger audiences with anti-smoking messages and support. Education and awareness campaigns on how to quit and encouraging people to visit health centres for support. Creating an environment that encourages smokers to seek help without fear of judgment can also help in promoting quitting. Programs that enhance recovery rates through

cessation support, counseling, and pharmacological aids are crucial for promoting a healthier population and reducing smoking prevalence.

4. Monitor and Evaluate Interventions

There is need for regular assessment of the implemented programs to monitor their impacts on public health. The initiatives to reduce smoking rates need to be visited regularly to identify successful strategies and areas for improvement. Moreover engagement in long term studies to assess the effectiveness of interventions on different demographic groups, particularly focusing on gender-specific behaviors is also important so as to reduce smoking related mortality.

5. If smoking becomes endemic $R_0 > 1$, it is necessary to have ongoing public health efforts to manage its prevalence. In this state, interventions must be sustained and adapted to changing demographics and behaviors.

6. Reduce of Mortality Rates ω_m, ω_f

While higher mortality rates may decrease smoking prevalence, there is urgent need for effective cessation strategies and health education to prevent smoking-related deaths. People affected with smoking related diseases need to seek help very early to avoid smoking related mortality.

7. Early Intervention Strategies

Implementing targeted campaigns during periods of rising smoking prevalence is critical to prevent the normalisation of smoking. It is critically important to implement these strategies as early as possible before any harm

8. Policy Development

There is need for legislation that increases taxes on tobacco products, restricts advertising, and mandates warning labels. Policies which encourages investment in research to further understand smoking dynamics and the effectiveness of different intervention strategies.

Here is a summary of policies that can be implemented to reduce smoking and its effects It is important to mainly focus on ways to reduce male rates since males ($R_{m0} = 0.188$) are dominating in the spread of smoking as compared to females ($R_{f0} = 0.00029$)

Intervention Target	Recommended Action	Expected Impact
reduce contact rates(β_m, β_f)	smoking bans, awareness campaigns	reduction in R_0
increase recovery rates(γ_m, γ_f)	educational campaigns and cessation strategies	reduction in R_0
reduce recruitment (σ_m, σ_f)	Youth prevention programs	Limits new smokers

5.4 Future works

Based on the current study's findings and limitations, future studies may focus on the following to enhance the model's realism, applicability, and public health impact:

1. Add Age-Structured Compartments

It is important to consider age based compartments to ensure effective smoking initiation and cessation since rates vary by age (e.g., teens vs. adults). Divide the compartments into age groups and divide parameters for adolescents due to peer pressure. This will help to identify critical age groups for intervention (e.g., school-based programs).

2. Model Time-Varying Interventions

In our current model parameters a constant, but parameter values can change due to new policies and may be change due to seasons. Therefore it is important to introduce time dependent parameters. This will help to optimise timing and duration of interventions.

3. Models with control measures

In our model measures to reduce smoking are not taken into consideration yet they have a great impact in reducing smoking. Therefore future research may focus on the effects of measures to control smoking such as educational campaigns and cessation measures on the dynamics of smoking models .

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